MIL-T-83235 (USAF)
17 November 1969
SUPERSEDING
MIL-T-27441C(USAF)
12 September 1968

MILITARY SPECIFICATION

TRUCK, FIRE FIGHTING A/S32P-4; AIR TRANSPORTABLE, MULTIPURPOSE

1. SCOPE

1.1 This specification covers one type of medium weight, diesel-engine driven, air transportable fire fighting truck, designated A/S32P-4.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

L-P-383	Plastic Material, Polyester Resin, Glass Fiber Base, Low Pressure Laminated
O-F-555	Foam Liquid, Fire Extinguishing, Mechanical
W-C-596	Connector, Plug, Electrical; Connector, Receptacle, Electrical
VV-F-800	Fuel Oil, Diesel
ZZ-H-451	Hose, Fire, Woven-Jacketed, Rubber- or Cambric-Lined, with Couplings
ZZ-H-561	Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water
GGG-A-926	Ax (Single Bit, Double Bit, Pick Head and Mattock Head)
GGG-B-101	Bar, Chisel, Pinch, Pry, Wrecking, Digging, Tamping, Digging and Tamping, and Crowbar
GGG-C-740	Cutter, Bolt
GGG-W-665	Wrench, Spanner
PPP-C-96	Can, Metal, 28 Gage and Lighter

Military

MIL-J-449	Jacket, Fire Hose (2-1/2 Inch)
MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-C-3432	Cable and Wire, Electrical (Power and Control, Flexible and Extra Flexible, 300 and 600 Volts)
MIL-F-3541	Fittings, Lubrication
MIL-G-3787	Glass, Laminated, Flat (Except Aircraft)
MIL-H-3992	Hose and Hose Assembly, Rubber, Air and Vacuum Brake, Automotive

FSC 4210

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MIL-H-4497	Hose, Cotton, Rubber Lined, Water, Braided Construction, with Couplings
MIL-H-4697	Heater, Coolant, Engine, Gasoline Burning, General Specification
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-C-7580	Coupling, Fire Hose, Reusable, High Pressure
MIL-E-7589	Extinguisher, Fire Vaporizing, Type D-2, Bromochloromethane, Portable (Extinguisher, Fire, Bromochloromethane, Portable, Type D-2)
MIL-H-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-B-8607	Belt, Safety, Passenger Type
MIL-H-10201	Horn, Fluid-Actuated, Horn, Electrically-Actuated, and Buzzer, Electrically-Actuated
MIL-S-11086	Strap, Hose, and Ladder
MIL-B-11188	Battery, Storage, Lead-Acid
MIL-R-12144	Reflector, Indicating, Clearance
MIL-S-12165	Strainer, Suction, Fire-Hose, and Strainers, Suction, Hose
MIL-L-12197	Ladders, Fire Extension; Ladders, Fire, Single; and Ladders, Fire, A-Folding Extension
MIL-N-12314	Nozzle, Fire Hose, Water, Spray and Stream
MIL-F-23905	Fire Extinguishing Agent, Light Water, Liquid Concentrate, (6 Percent)
MIL-C-25050	Color, Aeronautical Lights and Lighting Equipment, General Requirements for
MIL-N-26206	Nozzle, Fire Hose, Foam-Water, Aircraft Fire Fighting, 60 GPM and 100 GPM
MIL-L-26598	Lantern, Electric LEU-2/E
MIL-S-38265	Saw, Circular Blade, Metal-Cutting MXU-346/E, Gasoline-Engine-Driven, Portable
MIL-P-43116	Pike, Wire and Pikes, Pole
MIL-C-52404	Connections, Hose, Fire and Water

STANDARDS

Federal

595	Colors

Military

MIL-STD-12	Abbreviations for Use on Drawings, Specifications, Standards, and Technical Documents
MIL-STD-17	Mechanical Symbols for Aeronautical, Aerospacecraft and Spacecraft Use
MIL-STD-100	Engineering Drawing Practices

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MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-209	Slinging Eyes and Attachments for Lifting and Tying Down Heavy Military Equipment
MIL-STD-281	Automobile, Trucks, Truck-Tractors, Trailers and Trailer Dollies, Preservation and Packaging of
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	Maintainability Demonstration
MIL-STD-756	Reliability Prediction
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1400	Engine, Gasoline or Diesel, Methods of Test
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure
MS33586	Metals, Definition of Dissimilar
MS51335	Pintle Assembly, Towing - 18,000 Lbs Capacity, Manual Release
MS75058	Connector, Receptacle, Electrical-Slave, 2 Contact, 24 Volt, Waterproof

Air Force-Navy Aeronautical

AN2552	Receptacle - External Power, 28 Volt DC

DRAWING

Air Force

44B5077	Receptacle Assembly - External Power Source

PUBLICATION

Air Force Bulletin

518	Cargo Aircraft Compartment Dimensional Data

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Department of Transportation

Motor Carrier Safety Regulations

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D. C. 20591.)

National Fire Protection Association

NFPA Standard 412	Test Procedures for Aircraft Rescue and Fire Fighting Vehicles Utilizing Foam

(Applications for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110.)

Society of Automotive Engineers

SAE Standards and Recommended Practices

J639	Safety Practices for Mechanical Vapor Compression Refrigeration Equipment or Systems used to Cool Passenger Compartment of Motor Vehicles
J645	Automotive Transmission Terminology
J833	Male Physical Dimensions for Construction and Industrial Equipment Design
J898	Control Locations for Construction and Industrial Equipment Design
J899	Operator's Seat Dimensions - Construction and Industrial Equipment Design
J925	Minimum Access Dimensions - Construction and Industrial Machinery

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, New York 10001.)

United States of American Standards

S1.4	General Purpose Sound Level Meters
S1.6	Preferred Frequencies for Acoustical Measurements
Z24.10	Octave-Band Filter Set for the Analyses of Noise and Other Sounds

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(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The truck shall consist of the following major components:

Item or System Name	See Requirement
Basic truck chassis	3.7
Lighting	3.8
Emergency warning devices	3.9
Cab and body	3.10
Instruments and controls	3.11
Agent system	3.12
Auxiliary fire fighting equipment	3.13
Fire fighting kit, structural KMK-391/S32P-4	3.14
Winterization	3.15
Air conditioning kit	3.16.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials shall be specified herein. Materials not specifically covered by this or applicable specifications shall be suitable, in every respect, for the intended purpose. Unless otherwise approved by the procuring activity, wood shall not be used in vehicle construction.

3.4.1 Metals. Where practicable, the metals used throughout the construction of the truck shall be of high strength-weight ratio. Unless otherwise specified herein, the parts and components of the agent system and winterization distribution systems, except the engine, which come into contact with the fire fighting agents, water, foam, and foam concentrate, and coolant liquid shall be fabricated of copper, brass, bronze, or 300 series stainless steel. The cab and body shall be fabricated of aluminum.

3.4.2 Dissimilar metals. Insofar as practicable, the use of dissimilar metals in contact with each other shall be avoided. However, metal plating or metal spraying of dissimilar base metals to provide similar or suitable abutting surfaces will be permitted. The use of dissimilar metals separated by suitable insulating material will be permitted. Dissimilar metals are defined in MS33586.

3.4.3 Protective treatment. When materials are used that are subject to deterioration when exposed to climatic and environmental conditions likely to occur

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during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack, or scale with age or extremes of climatic conditions shall not be used.

3.4.4 Corrosion resistance. The materials used throughout the construction of the agent system shall be resistant to corrosive action of the extinguishing agent.

3.5 Design and construction. The truck shall be of a closed, cab forward design utilizing a 6 by 6 chassis. The cab shall provide seating space for not less than four people. The configuration shall provide for frontal approach to fires, and shall be smooth in outline with rounded corners and no protruding portions. The truck shall be equipped with a complete and well designed water-foam fire fighting system. The truck shall be designed in accordance with best engineering practice and shall incorporate the minimum number of controls and special features required to provide safe and foolproof operation of the truck and to meet the requirements specified herein. Mechanical controls shall be used wherever possible. All parts of the truck shall be of the latest design and of the best quality. The parts shall be of such size, material, and strength as to properly sustain the maximum allowable loads imposed upon them with a maximum working efficiency and minimum wear during operation. The truck shall be so constructed that no parts will work loose in service and all liquids including fire fighting agents, coolants, and lubricants will not spill or leak under any operational condition. They shall be built to withstand the strains, shocks, vibrations, and other detrimental conditions incident to operation, maintenance, shipping, and storage with minimum loss of time for maintenance, repair, and servicing.

3.5.1 Functional design. The truck shall be designed to operate efficiently on fixed and forward airbases world wide for the purpose of providing quickly responsive fire suppression and extinguishing capability of aircraft fires and real material and property fires, and rescue of entrapped personnel under all climatic conditions. The truck shall be designed to operate in conjunction with other equipment and at sites where establishment of an extensive complement of other fire fighting equipment and complete maintenance facilities are not warranted. The truck shall be capable of operating on paved surfaces and traversing cross-country terrain as required to accomplish the fire fighting mission in the area of the airbase.

3.5.2 Reliability. The truck shall have a predicted MTBF (mean-time-between-failures) of not less than 300 pumping hours. A prediction shall be made in accordance with the following provisions:

a. For reliability purposes, a failure shall be all malfunctions that could prevent satisfactory operation, within the performance limits specified herein, or safe operation in the primary function of fire fighting.

b. The prediction shall be based on the principles of the design prediction procedure of MIL-STD-756 on the assumption that random failures of a system can be predicted as a function of the random failure histories and estimates of the components.

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c. Preventive or scheduled maintenance actions to preclude wear-out or time-dependent failures shall not be included in the MTBF prediction. Data and experience indicating wear-out characteristics shall be used to develop recommendations for preventive maintenance. Such preventive maintenance actions and predicted malfunctions that would not prevent satisfactory operation (not failures for reliability purposes) shall be reflected in any required maintainability calculations.

d. Failure rate data and estimates used in the prediction shall be modified to account for relative times of component use and for stresses, loads, and environments expected compared to those of the data.

e. The intent of the prediction shall be not to require extensive component testing to develop component failure rates. The intent shall be to apply a prediction to each component based on engineering evaluation of components and designs in related environments and applications. Documented failure rate data used as the basis for predictions shall be available for each component.

f. Failure rate data based on dimensions other than pumping hours shall be converted to failures per pumping hour unless data and analyses acceptable to the procuring activity indicate other conversions: 4 miles equals 1 use cycle and 2 use cycles equals 1 pumping hour.

3.5.2.1 Reliability demonstrated. The MTBF of not less than 300 pumping hours shall be demonstrated by the tests specified herein.

3.5.2.2 Failure mode and effect analysis. A study and analysis shall be made in connection with the reliability prediction and with all maintenance and maintainability analyses to determine the effect of each potential mode of failure of each component. The following conditions shall apply:

a. The analysis shall be down to the level of at least each component normally replaced as a unit, or to each functional block or division, or to some logical combination of the two. Listing of each part (seals, lamps, etc) will not be required.

b. Each potential mode of failure shall be classified among at least the following classes:

(1) Hazardous - The failure or condition is capable of causing death or injury to personnel or of causing destruction of or damage to the equipment itself or to related equipment. The analysis for hazardous conditions shall include, in addition to failure modes, any hazards possible when the equipment is operating properly. Wherever the state-of-the-art makes it possible, hazards shall be eliminated by the design and construction. Wherever complete elimination of a hazard is not possible, the study shall document the problem and the actions taken to minimize the hazards.

(2) Mission loss - The failure can prevent satisfactory performance of primary functions and cannot normally be repaired at remote sites.

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(3) Mission delay - The failure can prevent satisfactory performance of primary functions but can normally be repaired at remote sites without the complete complement of tools and support equipment expected at conventional bases or airfields.

(4) Mission degradation - The malfunction does not permit satisfactory performance as fully as specified but would permit completing primary functions (loading, travelling, and pumping) at reduced efficiency or frequency.

(5) Maintenance event - The malfunction or event entails adjustment or maintenance but does not delay or prevent performance of primary functions.

c. Particular components designated for use in the equipment shall be included in the analysis and shall be applied to eliminate hazards and to minimize failure effects. Where modification of such components is required, the procuring activity shall be notified.

3.5.3 Maintainability design. The truck shall be designed in accordance with the design guidelines of MIL-STD-470, paragraph titled "Establish Maintainability Design Criteria." The truck shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with the performance specified herein.

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure.

c. Permit attaching operations, adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components by personnel wearing heavy gloves or mittens and bulky clothing and footwear. Locking devices, controls, and fasteners which can be easily overtorqued by operators lacking feel through heavy gloves or numbness shall not be used. Parts and components shall be located for ample and rapid access unless performance will be appreciably degraded by the accessible location. In performing maintenance, if engineering reasoning or data determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility.

d. Permit maintenance with general-purpose tools and equipment normally available commercially. Use of special-purpose tools and equipment shall be subject to approval by the procuring activity.

e. Minimum number of tools required for maintenance by design practices such as reducing the variety of bolthead sizes to the practicable minimum.

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f. Permit accomplishment of all servicing operations, all scheduled preventive maintenance (excluding major overhaul), and not less than 98 percent of all corrective maintenance tasks within the following downtimes:

(1) Mean corrective maintenance downtime shall be not more than 1 hour.

(2) Maximum corrective maintenance downtime shall be not more than 3 hours, except as specified in 3.7.2.2.

(3) Maximum total downtime to perform all preventive maintenance and servicing tasks shall be not more than 35 minutes.

3.5.3.1 Foolproofness. Where improper installation of a part could cause malfunction of that part or of the truck, the part shall be so designed that it can be installed only in its proper operating position.

3.5.3.2 Major assemblies. To the maximum practicable extent, it shall be possible to remove major assemblies without the necessity for draining liquids or coolants, or dismantling other assemblies. Major assemblies shall be equipped with lifting eyes or other suitable provision for ready attachment of lifting facilities.

3.5.3.3 Pilots or guides. Pilots, guides, slides, carriages, or other features shall be included where such provisions will add materially to the ease of removal and installation or attachment of components or parts.

3.5.3.4 Cover plates. Cover plates, which must be removed for component adjustment or for component or part removal, shall be equipped with substantial, quick-disconnect fastenings.

3.5.3.5 Detachable fittings. Disconnect plugs and receptacles, junction boxes, bus bars, multiple line connectors in the electrical system, and readily detachable and attachable type fittings in hydraulic and pneumatic systems, as applicable, shall be utilized to permit rapid component removal and replacement with all disconnect points clearly indicated.

3.5.3.6 Hourmeter. An hourmeter shall be provided to record hours of engine operation. The hourmeter shall be so mounted on the truck that it can be conveniently and easily read by personnel servicing the truck.

3.5.3.7 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with an adequate means for preventing loss of proper tightness and adjustment. When subject to removal, replacement, or adjustment, attaching hardware for any component shall be easily removable and shall not be swaged, peened, staked, or otherwise permanently deformed. Sheet metal screws shall not be used.

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3.5.3.8 Insulation. The insulation shall be fire resistant and shall be of a type that will not pack or settle. Any condensate formed between the walls shall be drained by gravity flow. The average heat loss per square foot per degree Fahrenheit shall not exceed .24 BTU per hour. All insulated areas exposed to abrasion or damage due to equipment storage and occupation by operating or maintenance personnel shall be protected by inner paneling.

3.5.4 Safety features. All space which is occupied or in which work is performed during operation, servicing, and maintenance of the truck shall be free from hazardous protrusions, edges, cracks, or other elements which might cause injury to personnel. Safety features, such as steps with antiskid treads, catwalks, handrails and guards, shall be provided at all points where the protection of personnel is in question. Except for the treads on the body top and the external means of access to the body top, all steps shall be enclosed and protected from the weather. A stairway, rather than ladder-type steps, shall be provided in any location where more than one step is involved. The height between steps shall be less than 20 inches. The lower cab step shall be less than 24 inches from the ground. The tread of the bottom steps shall be not less than 8 inches in width and succeeding steps not less than 16 inches in width. All steps shall be not less than 6 inches in depth. Entrance and exit from the cab shall not be hindered in any manner.

3.5.4.1 Catwalks. Catwalks which are constructed and supported to withstand the loads imposed by service requirements shall be furnished to provide access to all equipment mounted on the top of the truck. The catwalks shall be raised 1 inch above the surrounding surface, shall be at least 12 inches wide, and shall be constructed of expanded metal with nondirectional projection mesh.

3.5.4.2 Forward visibility. The visibility requirements specified herein shall be applicable to all eye heights within the range of small man (clothed) through large man (arctic clothed) as specified for seated body dimensions in SAE J833 while sitting in a fixed, normal operating position. The design and arrangement of the cab and components contained therein shall be such as to afford optimum visibility for control of the truck and performance of fire fighting operations. The driver and equipment operator shall have a lateral field of vision of at least 200° and the ground must be visible to the driver and the equipment operator at a point not greater than 15 feet from the truck through the full included angle of vision as shown on figure 1. The angle of visibility above the horizontal for both the driver and equipment operator through the specified horizontal angle of vision shall be at least 5°. The restriction of the horizontal angle of vision by obstructions, such as window frames, corner and door posts, etc, shall not exceed 7° each. Forward vision for both the driver and equipment operator through the windshield between the forward corner posts of the cab shall be unobstructed. A center post is permissible if the center post width is restricted to that width (in all cases under 2 inches) which imposes no blind area at a distance of 40 feet from the vehicle when viewed by either the driver or equipment operator using both eyes sitting in a fixed normal operating position.

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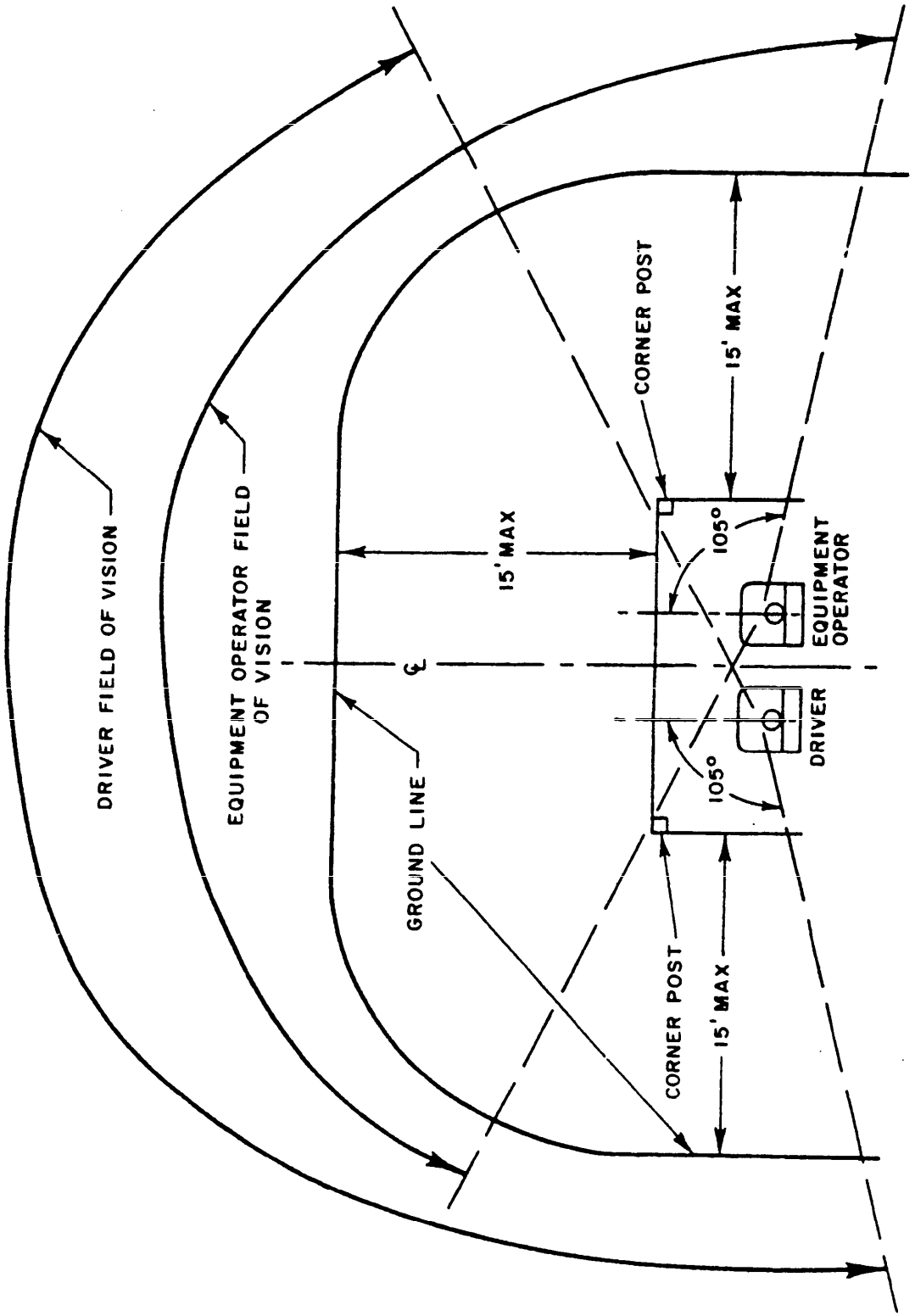


FIGURE 1. Horizontal Angles of Vision

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3.5.4.3 Component protection. All oil, hydraulic, and air system tubing, and all electrical lines shall be located in protected positions. They shall be properly clipped to the frame or body structure and, except where a through-frame connector is necessary, shall be furnished with metal protective looms or grommets at each point where they pass through panel or structural members.

3.5.4.4 Brush guards. All front grills, louvers, headlamps, tie rods, drive shafts, and other vulnerable components shall be protected by component locations or substantial guards to prevent damage to the components from brush, stones, logs, etc, likely to be encountered by the truck during off-road performance.

3.5.4.5 Hazard guards. All rotating or reciprocating parts and all parts subject to high operational temperatures or that are electrically energized, that are of such a nature or so located as to be or become a hazard to the safety of operating personnel shall be insulated, enclosed, or guarded.

3.5.5 Servicing provisions. Drains, filler plugs, grease fittings, hydraulic lines, bleeders, and check points for all components shall be so located that they are readily accessible and do not require special tools for proper servicing.

3.5.5.1 Filler caps. All filler caps shall be attached to the filler neck with a safety chain to prevent loss.

3.5.6 Lubrication. The truck shall be designed and constructed to operate satisfactorily when lubricated with standard lubricants as specified in MIL-M-8090. Wherever practicable, lubricated-for-life components or components not requiring lubrication shall be used in the construction of the truck. Grease and oil seals shall be designed and located to provide maximum accessibility for inspection, servicing, and replacement. Panels which must be opened for access to lubrication points shall be hinged.

3.5.6.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541 and shall be located in accessible, protected positions. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication, or which are likely to be overlooked because of inaccessibility.

3.5.7 Air transportability. The trucks shall be designed for air transportability (see 3.19 and 3.20) in accordance with MIL-A-8421, in a Type C-130 aircraft equipped with an A/A32H-4 dual-rail cargo handling system. Transportability shall be in accordance with Bulletin 518 and figures 2 through 5 herein for the rail system and vehicle loading conditions (see 6.4).

3.5.7.1 Component removal. Disconnects shall be provided for component removal and reinstallation to a fully operational condition (see 3.10 and 3.19) of the roof turret and other protrusions above the required height for air transport. If necessary to

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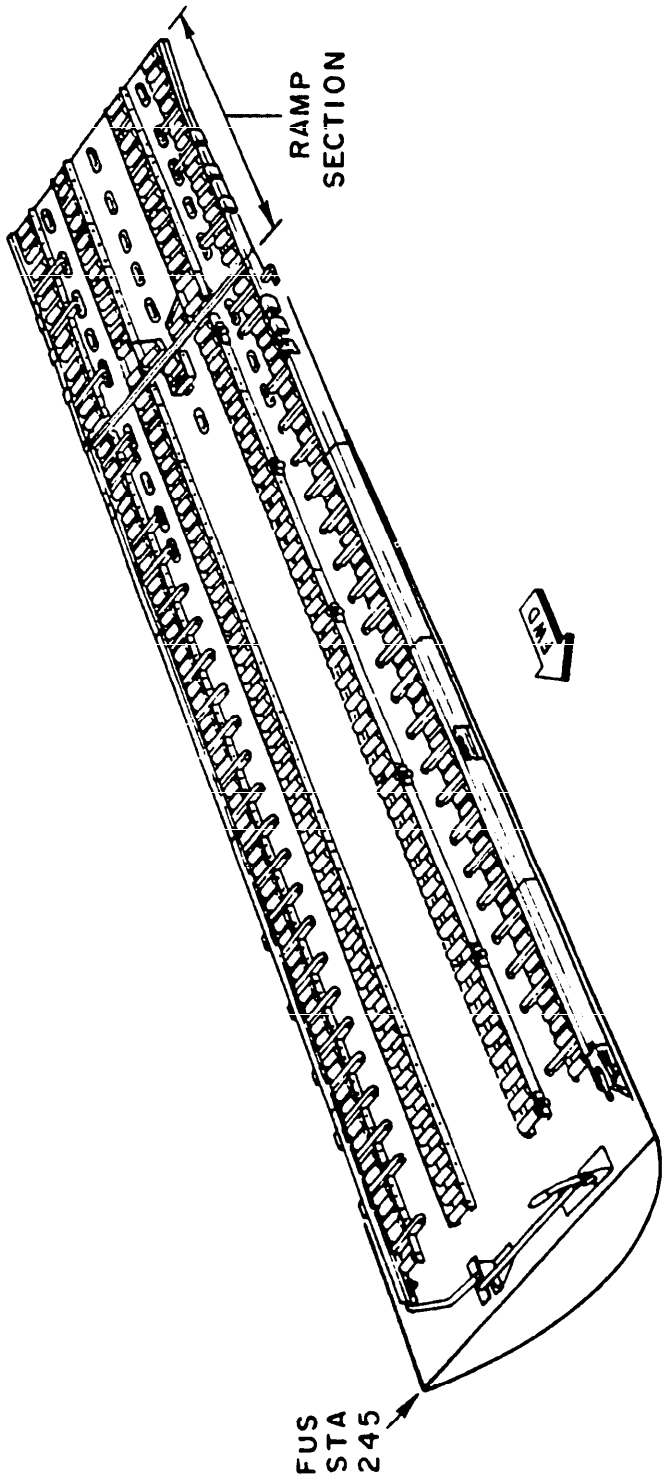


FIGURE 2. A/A32H-4 Dual-Rail Cargo Handling System

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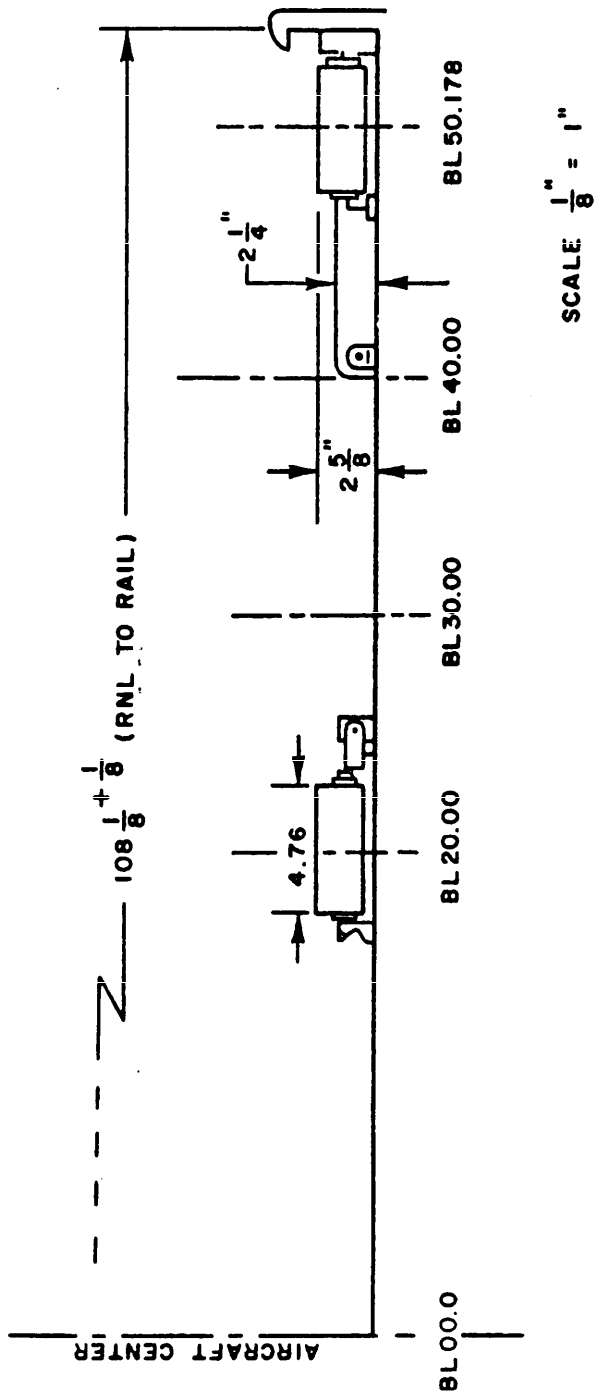
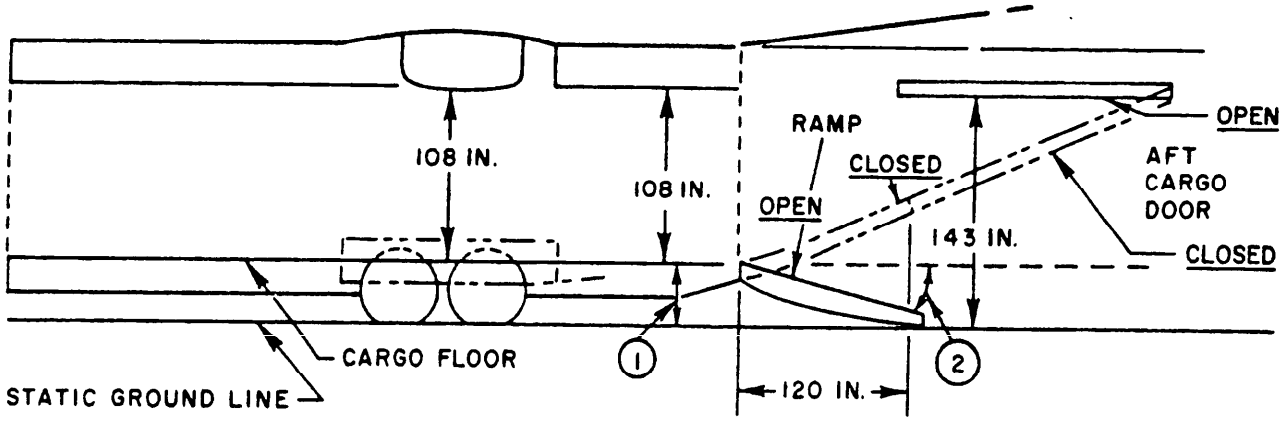


FIGURE 3. C-130 A/A32H-4 Cargo Handling System

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1. Floor height above ground. Variable 45 to 41 inches empty - 40 to 39 inches loaded

2. Ramp Angle, variable from 11.5° to 15°.

FIGURE 4. Cargo Compartment Dimensions (Rail System Not Included)

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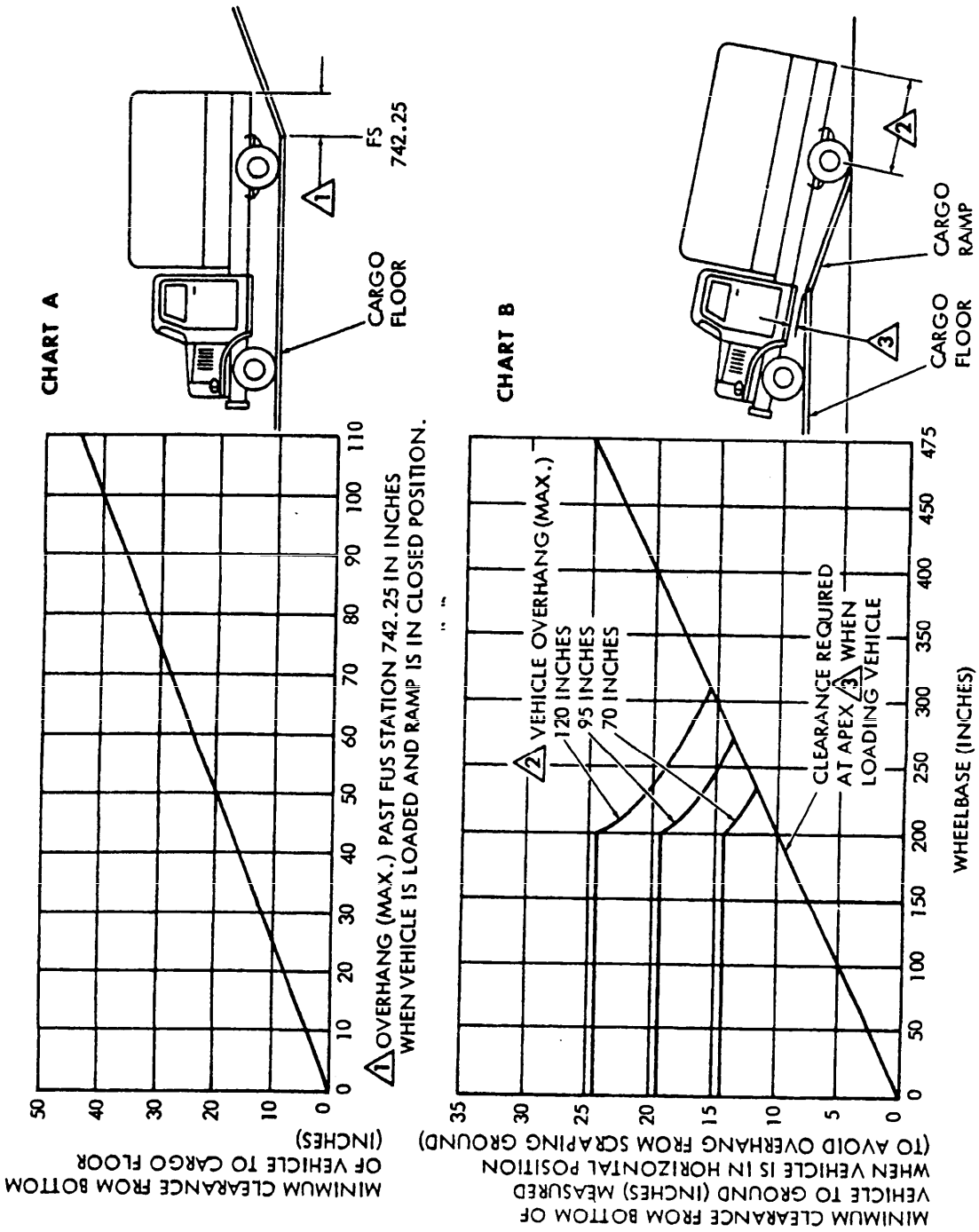


FIGURE 5. Vehicle Overhang and Projection Limits

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provide the required truck height for air transport, removal of the cab roof section will be permitted. Removal and reinstallation of the roof section and top mounted equipment shall be limited to removal or attachment by standard fasteners or fittings. The weight of individual parts or assemblies which must be handled or positioned during removal and reinstallation operations without mechanical assistance provided as part of the component removal kit (see 3.5.7.1.1) shall not exceed the limits specified in MIL-STD-1472 for two-man lift. Disassembly of parts or components which could be easily damaged or will require special fit-ups or protection during transport, assembly, or disassembly operations will not be permitted. Reassembly of truck components from the C-130 air transportable configuration to the fully operational configuration, including storage of the component removal kit, shall be completed by not more than three men in an elapsed time not to exceed 4 hours. During 2 hours of this 4-hour period, the truck shall be capable of immediate operational use to the extent described in 3.5.7.2. The disassembly of truck components from the operational to the air transportable configuration shall be subject to the 4-hour three-man limitation described above. All equipment and components removed for air transportability shall be stored on the truck. This storage shall not hinder or preclude operation of the vehicle as described in 3.5.7.2.

3.5.7.1.1 Component removal kit. Component or part removal shall be accomplished using equipment provided for this purpose in kit form. The kit equipment shall be storable on the truck during all modes of specified operation. All necessary holders, compartments, toolboxes, brackets, mounting pads, etc, for use and storage of the kit equipment shall be provided with the truck. The weight of the individual parts or assemblies of the kit which must be handled or positioned during truck component removal or reinstallation operations shall not exceed the limits specified in MIL-STD-1472 for two-man lift.

3.5.7.2 Limited operational capability. The truck and agent system, except the roof turret, shall be operational without degradation of performance or compromise of personnel utility of the components or assemblies not removed while the truck is in an air transport configuration. All necessary valves, line caps, or plugs shall be provided to permit this operation. A temporary canvas or other approved protective cover that can be stored on the truck in a convenient location when not in use shall be provided to permit use of the truck in air transport configuration.

3.5.7.3 Lifting and tiedown provisions. Lifting and tiedown eyes in accordance with MIL-STD-209 shall be provided on the truck frame. The truck shall be provided with sufficient attachment points so located that the truck can be restrained in all directions during air transport. The attachment points of the truck shall be designed for use with attachment points in the aircraft which, in the expanse of aircraft floor, are of 10,000 pounds capacity and placed in a 20-inch grid pattern with 25,000-pound attachment points on approximate 40-inch centers along both sides of the aircraft floor. The tiedown points on the truck shall be designed and located on the basis that restraining chains will be at an angle of 30° with the aircraft floor. Truck attachment point loading capacities shall be downgraded accordingly from the standpoint of restraining horizontal, vertical, and lateral g-loads on the vehicle.

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3.5.7.3.1 Clearance. Sufficient clearance shall be maintained around all attachment points to allow for the attachment of cables and chains without interference from any other structure on the truck. The tiedown points shall have a minimum clear opening of 4 inches to allow attachment of a chain tiedown with a grabhook.

3.5.8 Basic design. The truck shall be designed to provide the following basic features:

3.5.8.1 Weight distribution. The difference in load between the wheels on any axle shall be no more than 5 percent of the average wheel load on the axle for any truck loading condition. The difference in load between any two axles shall be no more than 10 percent of the average axle load for any truck loading condition.

3.5.8.2 Ground clearance. The ground clearance under the lowest projection of the fully loaded truck shall be not less than 12 inches.

3.5.8.3 Turning clearance. The turning clearance (over the outermost portion of the truck) shall be the minimum practicable, and the distance required to make a 180° turn in either direction shall not exceed 100 feet in width.

3.5.8.4 Riding qualities. The truck shall possess riding qualities that will permit operation over rough roads and adverse terrain without undue discomfort to operating personnel or damage to the truck or fire fighting equipment. Undue discomfort to operating personnel shall not be encountered where the truck is operated for periods as long as 8 hours.

3.5.8.5 Flexibility. The truck shall be so constructed that when fully equipped and loaded with its rated payload, with tire chains installed on all wheels and the front wheels cramped at any angle, blocks at least the height of the tire cross-section diameter can be placed under any wheel or any combination of two or more wheels without interference or part failure.

3.5.8.6 Waterproofing. All components shall be so designed and protected that their normal functioning will not be impaired by heavy rains, road splash, formation of condensation, spillage of extinguishing agents from nozzles and fittings, recharging operations, and leaks in the piping system.

3.5.8.7 Simplicity and ease of operation. All components and systems shall be so designed that all operations can be performed without exertion of effort by operating personnel in excess of the limits specified in MIL-STD-1472. Any fire fighting equipment controls located outside the truck shall be not higher than 66 inches above the ground or catwalk.

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3.6 Performance

3.6.1 Environmental conditions. The truck shall be capable of withstanding the following conditions without detrimental effect to subsequent operation:

a. Starting and operating in ambient temperatures ranging from -40° to +125°F

b. Storage at temperatures ranging from -80° to +160°F

c. Relative humidity up to 100 percent

d. Exposure to salt fog

e. Fungus growth as encountered in tropical climates

f. Dust particles as encountered in desert areas.

3.6.2 Mobility. The fully loaded and equipped truck shall be capable of meeting the following requirements:

a. Maintain a speed of 60 mph on a dry level concrete road, 35 mph on gravel roads, 20 mph on Belgian Block pavement and 20 mph on cross-country terrain

b. Accelerate from a complete stop to a speed of 60 mph in 60 seconds on a dry, level, concrete road

c. Operate on smooth, dry, level concrete at speeds from 1 mph to at least 10 mph while discharging agents from both turrets at rated capacity without interruption

d. Ascend a smooth, dry, concrete road having a 5-percent grade at a maintained speed of 25 mph

e. Ascend, stop, start, and continue ascending on a 40-percent grade at a speed up to at least 1 mph with extinguishing agents being discharged at maximum rated capacity from both turrets without interruption

f. Descend, stop, start, and continue descending on a 40-percent grade at speeds of 1 to 10 mph with extinguishing agents being discharged at maximum rated capacity from both turrets without interruption

g. Ascend and descend a 60-percent grade

h. Operate in both directions on a 20-percent side slope with extinguishing agents being discharged at maximum rated capacity from both turrets without interruption

i. Operate in water to a depth of 2 inches at a speed of 40 mph

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j. Be held and controlled by the service brakes on an incline of 60 percent, and be brought to a complete stop, under any load condition within 30 feet from a speed of 20 mph on a dry, hard, approximately level road, free from loose material. The service brakes shall be capable of full recovery within 3 minutes after a run down a 3-mile (average 10 percent) grade, with transmission in high gear, and road speed controlled by repeated application of the brakes not to exceed 30 mph. The service brakes shall be capable of full recovery within 3 minutes after operation in water to a depth of 2 inches at 40 mph. The hand (parking) brakes shall hold the fully loaded truck motionless when headed in either direction on a 40-percent grade

k. Traverse in one pass, at low speed, terrain consisting of unflooded fine grained dry to moist soils having a minimum cone index (CI) rating in the critical layer of no greater than 55 (see 6.3.2)

l. Climb a vertical wall at least 18 inches high and negotiate terrain which will deflect the opposite wheels of the truck in alternately contrary directions to the extent built into the wave course (frame twister) at Aberdeen Proving Ground, Maryland

m. Negotiate, in both directions, a 30° ramp connecting two horizontal surfaces. The ramp length shall be at least 1.2 times the wheel base of the truck. The truck shall be capable of being driven forward, or backed from one horizontal surface across the ramp toe, ascend or descend the ramp, cross the ramp crest and proceed on the other horizontal surface without any portion of the truck scraping any of the three surfaces.

3.6.3 Fire fighting performance. The truck shall be capable of discharging water and foam selectively from the roof turret at flow rates of at least 400 and 800 gpm, the bumper turret at a flow rate of at least 300 gpm, and from the handline at a flow rate of at least 50 gpm for water and 100 gpm for foam using foam concentrate in accordance with O-F-555 and MIL-F-23905. All specified foam flow rates are for unexpanded foam. The truck shall allow selective operation of the 400/800 gpm roof turret, bumper turret, and handline in any combination to assure that rated performance of each discharge nozzle is achieved within 3 seconds after any nozzle or combination of nozzles are turned on or off for both foam and water discharge. The truck shall be capable of this performance regardless of the level of water and foam concentrate in the tanks.

3.6.4 Maneuverability. Engagement of the fire fighting equipment shall be possible with the truck moving at high speed without interruption of tractive effort for any period longer than that normally involved in the shifting of the chassis transmission gears. Such engagement shall not result in a harsh reduction in truck speed that would create a safety hazard or damage components and equipment. When the fire equipment is engaged, it shall be possible to engage the vehicle drive and to maneuver as specified herein without any interruption of fire fighting operation

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or decrease in fire fighting equipment performance. The possibility of stalling the engine when engaging the drive shall not be aggravated by the conditions existing due to driving the agent pumps or other power driven components.

3.6.5 Structural pumper operation. The truck shall be capable of discharging water at a rate of 1,000 gpm at 150 psig when drafting at an approximate lift of 10 feet, but not less than 9 feet, through 20 feet of 5-inch hard suction hose without pump cavitation.

3.6.6 Dry pump operation. The water and foam concentrate pumps shall operate dry at maximum rated pump speed for 10 consecutive minutes without appreciable wear or sustaining any damage.

3.7 Basic truck chassis

3.7.1 Chassis frame. The chassis frame shall be of adequate strength to withstand the imposed loads and to maintain the necessary chassis alignment during the intended service and shall be properly reinforced for the attachment of bumpers, rear pintle, towing and lifting shackles, and tiedown rings or eyes.

3.7.1.1 Bumpers. The front bumper and rear bumperettes shall be substantially mounted and so installed on the chassis frames to protect the front and rear of the truck. Installation of the rear bumperettes shall not interfere with the use of the pintle, and rear platform space.

3.7.1.2 Rear pintle. A rear pintle conforming to MS51335-2 shall be installed and braced on the chassis frame at the rear of the truck. The pintle hook shall be mounted to permit an 18,000-pound drawbar pull without causing permanent distortion or damage to the frame and mounting structure.

3.7.1.3 Towing shackles or hooks. Towing shackles or hooks for the attachment of towing chains shall be substantially mounted on the chassis frame at the front of the truck.

3.7.2 Truck engine. The truck and agent pumps shall be driven by a single diesel engine having not less than six cylinders, and of a design suitable for the intended service. The engine shall not be adversely effected by long periods of idling and slow speed operation.

3.7.2.1 Capacity. The capacity of the engine shall be such that under all specified conditions, the total horsepower required by all items driven by the engine will not exceed the intermittent brake horsepower rating of the engine as defined in MIL-STD-1400.

3.7.2.2 Engine installation. The engine shall be so located in a compartment as to provide maximum accessibility for servicing and maintenance from both sides of

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the compartment and in front, top, and beneath the engine without removing components other than underneath cowling and by opening compartment doors or the engine hood. The engine shall be so mounted and equipped with the necessary quick disconnects that engine-transmission packages can be removed or reinstalled by mechanics in less than 2 hours elapsed time.

3.7.2.3 Engine construction. The engine shall be complete with all accessories and auxiliaries necessary for efficient and dependable operation under the operating conditions specified herein.

3.7.2.4 Fuel. The engine shall operate satisfactorily and demonstrate the performance characteristics specified herein when using fuel conforming to VV-F-800.

3.7.2.5 Governor. An overspeed governor, accurate within 5 percent, shall be provided to limit the truck engine speed for all operations to the engine manufacturer's recommended top speed. If required, the engine shall be equipped with a separate governor to automatically govern the engine speed to that speed required to maintain fire pump discharge pressure during crash-rescue pumping operations. During structural pumping operations, the engine speed required to maintain pump discharge pressure shall be controlled by the hand throttle on the pump panel and shall not be governed by the separate governor.

3.7.3 Cooling system. The cooling system shall be of the pressurized type which utilizes a liquid coolant and circulating pump, with bypass to permit coolant circulation in the engine water jacket, when the thermostat is closed, until normal operating temperature is reached. The engine cooling system shall be thermostatically controlled to assure maximum engine performance when operating under the intended service conditions.

3.7.3.1 Temperature differential. With the truck operating in low gear at 70 percent full load measured on a vehicle dynamometer, in an ambient air temperature not lower than 60°F, or by commensurate road test in an air velocity not exceeding 3 mph (independent of air velocity produced by the radiator fan), the system shall be capable of maintaining a differential in temperature of not more than 95°F between the radiator inlet (engine outlet) coolant temperature and the ambient air temperature (with thermostat locked open).

3.7.3.2 Radiator. The radiator core shall be of materials and construction in accordance with the best commercial practice, and shall be so mounted that it will not be damaged by vibration or frame distortion. Expansion tanks of sufficient capacity shall be provided for the expansion of the coolant in the engine water jacket, the radiator, any heat exchanger, and any heating system provided. Provisions for air separation shall be included. Counterflow of heated air to the radiator from the engine compartment shall be effectively prevented.

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3.7.3.3 Fan. The design, mounting, and drive of the radiator fan shall be consistent with the sturdiness, efficiency, and quietness required by the intended service conditions.

3.7.3.4 Controlled louvers. Thermostatically controlled, adjustable louvers shall be provided which will prevent the flow of air through the engine compartment when the engine is cold, and allow free airflow when the engine is warm. Manual controls shall also be provided to permit manual opening of the louvers in the event of failure of the automatic controls. In the event the automatically controlled louvers are located where they can be exposed to adverse weather or wind-borne sand, fixed louvers shall be installed in conjunction with the adjustable louvers to protect the adjustable louvers from such environment.

3.7.4 Engine lubrication system. The lubrication system shall be of the oil pressure type and shall function satisfactorily on side slopes up to 20 percent, and on grades up to 60 percent.

3.7.4.1 Oil filter. An oil filter of the engine manufacturer's standard type and size for the engine provided shall be included. A filter bypass shall be provided.

3.7.4.2 Oil filler and dipstick. An oil filler for the crankcase shall be included. It shall be so constructed that oil can be readily poured from a 5-quart container, conforming to type III of PPP-C-96, into the filler opening without requiring the use of a funnel. A bayonet gage graduated to indicate a full and safe operating oil level shall be so located that its finger loop is readily accessible.

3.7.5 Engine starting. The engine shall be capable of starting within 15 seconds regardless of prevailing ambient temperatures within the range of +125° to -40°F. Installed glow plugs, fluid starting aids, and heat from the winterization system may be used to facilitate engine starting under the following conditions:

	Temperature Range (Degrees F)	Starting Aids Permitted
a.	40 thru 125	None
b.	0 thru 39	Glow plugs and fluid starting aids
c.	-40 thru -1	Glow plugs, fluid starting aids, and heat from the winterization system.

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3.7.6 Fuel system. The engine fuel system shall function satisfactorily on side slopes up to 20 percent, and on longitudinal slopes up to 60 percent. The system shall include at least the following components:

3.7.6.1 Fuel tank. The fuel tank shall have sufficient capacity to provide consecutively 100 miles of road operation at an average speed of 45 mph with a maximum payload, 8 hours standby at -40°F with combustion heaters maintaining the unit at operational temperature, and discharge of all extinguishing agents without refueling. Baffles shall be provided and secured in place. Sheet metal for the tank shall be terneplate or other suitable material with seams of welded construction, so formed that the terneplate finish on the interior will not be affected by the weld. The tank shall be so mounted that it will not be damaged by frame distortion or collision with roadway obstructions, and shall not be affected by heat or flames that may surround the vehicle or heat from the exhaust system. Suitable venting shall be provided. The fuel fill point shall be so located as to preclude fire hazards during filling operations and be accessible to personnel standing on the ground.

3.7.6.2 Fuel filter. A filter shall be so located in the fuel line between the fuel tank and fuel pump that it can be readily serviced without removing surrounding equipment. Unless the filter is mounted at a point higher than the fuel tank, a shutoff valve shall be included in the inlet line to the filter.

3.7.6.3 Fuel lines. Fuel lines of adequate size solid wall tubing shall be equipped with long nut compression type fittings. Each fuel consuming component shall have its own individual fuel line from the fuel tank.

3.7.6.4 Flexible fuel lines. Flexible hose shall be provided from the fuel line on the frame to the fuel system or the engine and at all points where vibration or flexure might cause tubing failure.

3.7.6.5 Injectors. The injectors shall be of the engine manufacturer's standard type and size for the engine provided.

3.7.6.6 Fuel transfer pump. An electric fuel transfer pump shall be provided and shall be so installed that it can be easily removed.

3.7.6.7 Air cleaner. A heavy duty air cleaner shall be furnished.

3.7.7 Exhaust system. The exhaust system shall be of substantial leakproof construction, amply protected, and securely mounted. Nuts used on exhaust fittings shall be noncorrosive materials. The exhaust pipe shall be so installed that it will not be damaged by vibrations.

3.7.7.1 Muffler. The muffler shall be so designed that it will effectively reduce exhaust noise and not create excessive back-pressure.

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3.7.7.2 Tailpipe. The tailpipe shall be located to the rear of the truck and shall exhaust the gases in such a manner as not to contaminate any area within or surrounding the truck, create a safety hazard, or damage any components. Provisions shall be incorporated to prevent water from heavy rains and use of the agent system from entering the tailpipe.

3.7.8 Power transmission system. The parts and components of the power transmission system shall have a torque capacity of not less than the maximum (gross) torque of the engine as developed through the lowest intervening gear reductions. The power transmission system shall transmit the power from the engine to the axles and the fire pump drive through an efficient set of torque or speed converters to produce the tractive effort, speed, and power required to meet the vehicular and pump performance specified herein. The truck and pump drive shall be so designed that flexibility of vehicle speed and tractive effort are maintained after pump engagement is made. After pump engagement, control of truck movement by the brake system alone will not be permitted.

3.7.8.1 Transmission and transfer case. The design of the transmission and transfer case shall provide for maximum vehicular performance and for ease and quietness of operation. The transmission shall be of a hydrodynamic, semi-automatic type as defined in SAE J645. If required to meet tractive effort demands of all conditions of service, a transfer case containing low- and high-range gear trains shall be provided. These units shall be of a constant mesh or synchronized countershaft design. All shifting of gears shall be possible under full power.

3.7.8.2 Transfer case output shafts. The transfer case drive for the output shafts shall contain features which will compensate for differences in speeds between the front and rear axles. Differential locking mechanisms shall be incorporated into the transfer case design.

3.7.8.3 Universal joints. Universal joints shall be of the latest design and shall be of identical make and type with metal construction throughout. They shall function satisfactorily under continuous operation at angles that will prevail with a fully equipped and loaded truck during all operation conditions.

3.7.9 Axles and tandem axle assemblies

3.7.9.1 Axles. The torque capacity of the axles shall be sufficient to slip the wheels under all possible service conditions including a 1.0 coefficient of friction. The axles shall be of such design and construction that, in the event of axle shaft failure at any point, the wheel will not be released from the truck. Interchangeability between front and rear axle components shall be maintained for components performing identical functions. Rear axles shall be identical. Axle ratings shall be not less than the imposed GVW load measured at the ground.

3.7.9.1.1 Axles and wheel mountings. The front and rear axle (nominal) treads shall be approximately equal and consistent with proper mobility and steering designs. The design shall include features to provide for easy removal of the wheels.

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3.7.9.1.2 Front axle. The front axle shall be equipped with steering drive ends which afford constant velocity.

3.7.9.1.3 Steering tie rod. The steering tie rod shall be so located that it will be adequately protected from damage. The rod shall be so threaded that fine toe-in adjustment can be accomplished.

3.7.9.1.4 Steering pivots. The steering pivots shall be equipped with bearings and with effective means of adjustment.

3.7.9.1.5 Cramping angle. The cramping angle of the front wheels shall be not less than 28°. Axle stops shall be provided which will accurately limit the cramping angle to the maximum angle intended.

3.7.9.1.6 Differentials. All axle assemblies shall be equipped with differentials. The differentials shall be of the same size in all axle assemblies and shall provide maximum interchangeability between all axle assemblies. Driver controlled differential lockouts shall be provided in all differentials.

3.7.9.2 Tandem axle assembly. The tandem axle assembly shall be provided with a means permitting differential action between the two axles with manually controlled lockout. The lockout control shall be located in the cab. The tandem axle assembly may be any of the following types:

a. Independently sprung wheels with interlocking fore and aft torsion bars

b. Spring-mounted platform with walking beams

c. Fully articulated, constant parallelogram, balanced spring suspension type, complete with axles, springs, torque rods, and other necessary parts. Lateral float shall not exceed 1/4 inch on either axle assembly.

3.7.10 Suspension system. The suspension system shall be such as to reduce the changes in acceleration and amplitudes of transverse and vertical vibrations incident to travel over open country and rough roads to provide the riding qualities specified herein. The design shall be such that the natural frequency of vertical vibration about any transverse axle will be less than 2.5 Hz. Any free vibration of maximum amplitude shall be effectively damped out in two cycles.

3.7.10.1 Springs. The springs shall have adequate capacity to transport the imposed loads under the intended service conditions. The springs shall be of such construction and material to insure maximum service life during cross-country operation while providing the required riding comfort.

3.7.10.2 Shock absorbers. When necessary, double-acting hydraulic type shock absorbers with adequate capacity to provide vehicle riding qualities consistent with the service requirements shall be furnished.

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3.7.11 Wheels and tires. The truck shall be equipped with single tires on all the wheels. Front and rear wheels and tires shall be of identical design.

3.7.11.1 Wheels. Wheels shall be 22.5 by 14.00 disc type with 10 holes on a 11-1/4 bolt circle.

3.7.11.2 Tires. The truck shall be equipped with 18 by 22.5, 16-ply tubeless tires with nondirectional cross-country tread.

3.7.11.3 Balancing. Each hub and brake drum shall be dynamically balanced. Each wheel and tire assembly shall be dynamically balanced separately.

3.7.11.4 Spare wheel and tire assembly. When specified (see 6.2), a spare assembly consisting of the proper type and size wheel and tire shall be provided with but not mounted on the truck.

3.7.12 Braking system. The braking system shall safely control the fully equipped and loaded truck under any operating condition, and shall comply with the braking and holding requirements of 3.6.2. The brake mechanism shall be readily accessible for external adjustment. If an emergency occurs, the service braking system shall be adequate to control the vehicle without vehicle air pressure.

3.7.12.1 Service brakes. The service brakes shall be of the hydraulic application type, with air boost, with master cylinders of compensating type having self-contained fluid reservoirs. A system shall be provided which incorporates two master cylinders and air boosts, one set for the front axle brakes and one set for the rear axle brakes. Air filters shall prevent brake system contamination. Brake operation shall not be subject to excessive fading. Brakes shall be of internal expanding, two-shoe type, and shall include dirt-tight enclosure plates and slack adjusters. Brakes, drums, and actuating cylinders shall be provided for each wheel. Drums shall have flanges or ribbing that will prevent objectionable distortion when brakes are applied. Brake drums shall be demountable without disturbing wheel bearings.

3.7.12.1.1 Brake system air compressor. The air compressor shall be of such design that the head and air lines to the air reservoir will be free of pressure when the compressor is not in operation. The capacity shall be sufficient for the intended service but shall be not less than 12 cfm. The compressor intake shall be properly connected to an air cleaner. An alcohol vaporizing air bath, with features to prevent overfill shall be provided for proper winterization.

3.7.12.1.2 Brake system air storage reservoirs. Adequate air storage reservoirs with buzzer-type low-pressure indicator shall be provided. The system shall be of the quick-pressure build-up type with the air compressor connected to a small reservoir which is pressure-relieved into the larger reservoirs when the minimum pressure is reached for optimum operation of the brake system. The time required

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to reach this pressure from zero pressure at engine idle speed shall not exceed 15 seconds. If other than brake air supply requirements are to be supplied from the reservoirs, a priority system shall be installed to assure that an adequate supply of air is always available to operate the brake system.

3.7.12.1.3 Air lines. The air lines shall be winterized to prevent freezing of the braking system. Brake flexible hose shall conform to MIL-H-3992.

3.7.12.2 Parking brakes. Parking brakes, independent from the hydraulic system of the service brakes, shall be of the hand lever operated type. They shall be so mounted that they can be effective at all times. The operating controls of the hand (parking) brakes shall be independent of the operating controls of the service brakes.

3.7.13 Steering mechanism. An efficient and substantial mechanical steering mechanism shall be located on the left side of the truck. The design shall permit convenient adjustment for lost motion. Not more than three turns of the steering wheel shall be required to turn the truck wheels from maximum cramping angle right to maximum cramping angle left or vice versa.

3.7.13.1 Steering wheel. The steering wheel shall be of such size and design that easy steering will be assured under all operating conditions. The diameter of the wheel shall be not less than 20 inches. Not less than 4 inches clearance shall be provided between the outer edge of the steering wheels and the nearest projection in the plane of the steering wheel. At least 2-1/2 inches clearance shall be provided to the nearest projection below the steering wheel. The steering column shall include a supporting bracket for proper attachment to the vehicle to prevent movement and vibration.

3.7.13.2 Steering power assist. A power assist system shall be provided which will permit the driver to turn the truck wheels on dry level concrete while the truck is stationary and the engine is idling with not more than 15 pounds force at the steering wheel ring. Manual steering shall be available in the event of power assist failure.

3.7.14 Electrical system. The truck shall be equipped with a complete 24-V electrical system. The components shall be so installed as not to be adversely affected by heat from the engine or exhaust systems. The engine switch shall not require a key for operation. A master switch which can be utilized to disconnect the batteries shall be provided. A light which will indicate any short circuits shall be provided. The electrical system shall provide full power requirements for operation of the truck and all special equipment and accessories.

3.7.14.1 Starter. An electric starting motor and engaging mechanism with adequate capacity for the intended service shall be provided. It shall be actuated by an electric solenoid and engaged by a switch on the instrument panel in the cab. A vacuum actuated safety device, or an equivalent approved by the procuring activity, shall prevent engagement of the starter drive when the engine is running.

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3.7.14.2 Truck alternator. The truck alternator shall be furnished complete with a rectifier and voltage regulation system capable of supplying at least 28V, 100 amp, dc power. The system shall be capable of suppling full power requirements for operation of the truck and all equipment and accessories under all specified conditions. The design shall be such that at least 25 percent of the normally connected load in addition to the winterization system can be carried by the alternator when the engine is idling, and satisfactory operation can be assured over the speed range which the engine is capable of attaining. The installation of the voltage regulator and reverse current relay shall permit operation of the winterization system when the truck engine is shut off and the battery charge is being maintained from an external 110V, 20 amp, ac source. The alternator drive shall be of a capacity adequate to transmit the power required under all service conditions. If a selenium plate rectifier is provided, the rectifier installation shall be such that sufficient airflow is provided at all engine speeds to keep the rectifier plates below a maximum temperature of 180°F during all extremes of ambient temperature and under any conditions of engine operation.

3.7.14.3 Batteries. Batteries shall be either BB-248/U (6TN) or BB-223/U (8T) conforming to MIL-B-11188. Batteries shall be so arranged and connected as to provide sufficient capacity to assure engine start during adverse cranking conditions for all extremes of operation specified herein. The batteries shall be so installed that they can be easily maintained, will operate at suitable temperatures, and will be adequately ventilated. The batteries shall be mounted in an acid-resistant tray. A sliding carriage shall be provided when a sidewise motion is necessary for either battery servicing or removal operations.

3.7.14.4 Slave and power receptacles. Two slave receptacles shall be installed at the rear of the truck and connected in parallel with the batteries. One slave receptacle shall conform to Drawing 44B5077 and AN2552 and shall be provided with a spring-loaded cover. The other slave receptacle shall conform to MS75058. Conductors shall be sized to allow for starting of the engine.

3.7.14.5 Wiring. The wiring shall be installed and marked in accordance with applicable portions of MIL-W-5088. Wiring shall be assembled into individual harnesses and cables extending from the various components to subpanels where all component wiring pertaining to a given system is to be integrated. It shall then be passed through a main cable for such system to the instrument panel controls and instruments pertaining to that system. Particular attention shall be given to waterproofing of the wiring system (see 3.5.8.6). Wiring shall have thermoplastic insulation and conform to SAE standards for low tension insulated cable.

3.7.14.6 Protective devices. Circuit breakers shall be installed, in lieu of fuses, in all circuits as required to properly protect the electrical systems. Wires in metal conduits shall be thoroughly insulated from such conduit. All circuit breakers shall be located in the cab and connected in an organized manner to facilitate determination of malfunctions or shorts.

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3.8 Lighting. The truck shall be equipped with clearance lights, parking lights, stop and tail lights, and turn signal lights in accordance with current Motor Carrier Safety regulations. Lenses shall be of heat-resistant material to prevent damage when near a fire. All lights shall be complete and of 24-V operating potential.

3.8.1 Headlamps. Sealed-beam-type headlamps shall be furnished. A means shall be provided for adjusting headlamp beams. Headlamps shall be mounted as low as practicable, and shall be so adjusted that the center of the upper beam will strike the ground 300 feet ahead of the fully equipped and loaded truck when standing on level ground.

3.8.1.1 Headlamp dimmer switch. The headlamp dimmer switch shall be of a button type mounted on the floor of the cab.

3.8.2 Front spot and illuminator lights. One 24-V spotlight, with lamp conforming to USA part No. 4577, shall be installed on each side of the truck at the top front of the cab. The spotlights shall be so designed that universal movement of the lamp housings is accomplished through operation of controls extended inside the cab and readily accessible one to the driver and one to the equipment operator. The controls shall permit movement of the lamp housings through horizontal included angles of at least 270°, and a vertical angle of at least 90° above and 35° below horizontal. On-off control switches shall be located inside the cab and on the extended controls. All mechanisms required to provide universal movement of the lamp housing shall be lubricated, and completely enclosed and sealed to prevent leakage. In addition to the spotlights, a wide angle, high intensity, iodine quartz, illuminator light shall be installed in the front center of the truck just below the windshield. The light shall be connected to a switch on the instrument panel.

3.8.3 Rear spotlights. One 24-V spotlight, with lamp conforming to USA part No. 4577, shall be mounted on each side of the top rear of the truck. The lights shall be connected to switches on the instrument panel and at the rear of the truck and shall be suitable for backup lights. The spotlights shall be so installed that movement controls can be operated from the ground by remote control. Except for the switches and controls, the design shall be identical to the front spotlights.

3.8.4 Interior lighting. The cab and all compartments shall be provided with adequate lighting to enable personnel to readily obtain the needed equipment and to make any adjustments that may be necessary during operations. Control switches shall be so installed that the compartment and cab light is automatically on when the door is open, and off when the door is closed. Additional manual switches shall be provided for control of the cab lights.

3.8.5 Trouble lights. A 24-V trouble light with reflector and self-rewinding reel shall be installed in a location convenient for use all around the vehicle. The light cable shall be of winterized material and of such length as to permit use of the light all around the vehicle. The lamp socket shall be of watertight design and shall contain a control switch.

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3.9 Emergency warning devices. The trucks shall be equipped with the following warning devices:

3.9.1 Horn. A horn conforming to MIL-H-10201, type I, style 2, shall be properly installed in a protected position. It shall be remotely actuated by a horn button or horn ring in the center of the steering wheel.

3.9.2 Siren-loudspeaker. A commercial 24-V combination electronic siren and loud speaking system shall be provided. The system shall have a minimum 75-watt output and shall be compatible with all current US military installed radio systems using either carbon or magnetic microphones. The operating switches and controls shall be so mounted and located as to be easily accessible to the driver and equipment operator in their normal sitting positions. The speaker shall be so designed and housed that continued operation under icing conditions is assured.

3.9.3 Rotating beacon. A 24-V warning beacon shall be installed on the top of the truck. The color shall be aviation red as specified in MIL-C-25050. A switch shall be provided on the instrument panel to control the light. The location of the light shall provide unobstructed view of the beacon to alighting aircraft and to personnel standing on the ground at a distance of not more than 25 feet around the truck (360°) and shall meet the following characteristics:

a. The beacon shall be of the rotating type with a flash rate of 60 to 100 flashes per minute.

b. A minimum light intensity in all vertical planes, expressed in terms of effective candlepower as determined by the following formula, equal to or exceeding the values specified:

$$I_E = \frac{\int_{t_1}^{t_2} I_{(t)}\, dt}{0.2 + (t_2 - t_1)}$$

Where I_E = Effective candlepower

$I_{(t)}$ = Instantaneous candlepower as a function of time

$(t_2 - t_1)$ = Flash time interval in seconds when t_2 and t_1 are obtained by experimental means to be those times when the effective candlepower (I_E) would be equal to the instantaneous candlepower at both t_2 and t_1.

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Light intensities (with red color filters)

Angle Above or Below Horizontal (in Degrees)	Minimum Effective Candlepower Red Light Required
5 above to 5 below	100
5 to 10 above	60
10 to 20 above	20
20 to 30 above	10.

3.10 Cab and body. The cab shall be heated and ventilated. Equipment and components which must be protected against climatic extremes shall either be enclosed in heated and insulated compartments or the equipment itself shall be heated and insulated. The cab floor to ceiling height shall be not less than 62 inches. The ceiling area above the seats and above the passage area required to reach and leave each seat shall be unobstructed. Not less than 12 inches shall be provided between the seated occupants head and the nearest ceiling obstruction. Seated body dimensions shall be as specified in SAE J833 for all personnel falling within the range of a small man (clothed) through a large man (arctic clothed) and allowing 2 inches for added height due to fire protective clothing. Piping and other components projecting inside the cab shall be padded. There shall be no hazardous protrusions which might cause injury to seated personnel due to lurching or other motion of the truck. An escape hatch shall be provided in the cab roof. The size of the hatch shall be not less than the recommended minimum dimensions for body access, 95th percentile, for arctic clothed personnel as specified in SAE J925. The hatch shall be adequately sealed to prevent the entrance of water during heavy downpours. Unless otherwise approved by the procuring activity, access provisions to the escape hatch shall conform to the requirements specified in 3.5.4. If cab roof removal is required (see 3.5.7), the design of the cab roof removable sections shall include sealing provisions to insure that the assembled cab will meet the performance requirements specified herein after being subjected to 25 complete assembly and disassembly cycles.

3.10.1 Seating. Comfortable seats designed in accordance with the dimensions specified in SAE J899 as normal and desirable shall be furnished for each of four personnel. The seats for the driver and equipment operator shall be independently adjustable both longitudinally and vertically. All seats shall be upholstered in accordance with best commercial practice for transport vehicles. The seats shall be located and arranged to provide maximum visibility through transparent areas, maximum use of controls, and direct unobstructed access for entrance and exit of personnel to and from their respective seats by means of the cab steps.

3.10.2 Grab bars. Grab bars shall be furnished to provide stability for crew members during rough, cross-country operations.

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3.10.3 Seat belts. Seat belts conforming to MIL-B-8607 shall be provided for all personnel and firmly anchored to the cab floor. Where necessary, the belt length may be extended by bracket or cable to accomplish mounting to the cab floor or cab frame members.

3.10.4 Radio mount. A space approximately 1-1/2 feet long by 1 foot wide by 1 foot high shall be provided within the cab for mounting the radio unit furnished for base operations. The location shall be such that the radio controls will be accessible to both the driver and equipment operator while sitting in their respective seats.

3.10.5 Doors. All outside cab and compartment door handles shall consist of a large loop at least 4-1/2 inches long, with the latch protected to prevent entrance of ice and rain into the mechanism. The outside door handles shall be not more than 66 inches above ground level and shall not protrude appreciably beyond the body skin. The inside door handles shall be not less than 27 inches above cab floor level. Weather stripping shall be provided to prevent drafts around the doors. The door latches and hardware shall be of quality design. If piano hinges are used, they shall be of stainless steel. Features shall be furnished which are capable of protecting the doors against winds and fast stops while in the open position.

3.10.5.1 Cab doors. The cab shall have at least one door on each side, so located and sized as to provide for easy entrance and exit of operating personnel and crew members. Handhold bars at least 24 inches long shall be provided at each door for assistance of personnel entering the cab. The design shall provide features which will prevent the doors from becoming damaged under any condition of vehicle operation required herein. All cab doors shall be so designed that they can be opened and closed from the inside and outside. If split cab doors are used, a safety guard shall be installed over the center hinge.

3.10.5.2 Compartment doors. Rear-hinged and bottom-hinged doors shall be avoided where possible. Top-hinged compartment doors shall be held in the open and closed positions by spring-loaded, telescoping support arms, and latches shall not be provided. The doors shall be easily opened or closed and shall not rattle during travel. The doors shall be counterbalanced where necessary by means of springs or shall be divided horizontally, if required, to facilitate easy and safe handling by men of short stature.

3.10.6 Windshield. The windshield shall be of laminated, semitempered, safety plate glass conforming to MIL-G-3787. The width of the windshield shall be the maximum consistent with the truck width. If necessary to meet specified winterization requirements, readily removable transparent panels of a size to fully match the windshield may be provided to direct hot airflow between the panels for defrosting and anti-icing. If glass panels are provided, they shall be safety glass.

3.10.7 Windows. Windows of laminated, semitempered, safety plate glass shall be provided in the cab and doors. All windows shall be of such size as to provide

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maximum practical visibility. Unless otherwise approved by the procuring activity, all windows, except the windshield, shall be capable of sliding open and shall be of double-pane dehumidified design or shall be provided with frost shields for winterization.

3.10.8 Floors. The cab floor and all compartment floors shall be constructed to withstand the loads imposed by the service requirements and shall be insulated where required to reduce heat loss. Floors that must carry the weight of personnel during operation shall have a nonskid surface and shall be of adequate strength to withstand service without permanent deformation. The floor in all access passages within the cab shall be flat and free from obstructions.

3.10.9 Body top. The top or roof of the body shall be designed and constructed to provide sufficient strength and rigidity to sustain the loads imposed by the stowed equipment and the personnel required to service the equipment.

3.10.10 Drain troughs. Troughs shall be provided on the truck to carry away all water and foam without spilling over or running into cab windows or over the windshield. The troughs shall not drain onto locations anticipated to be occupied by personnel during normal operations.

3.10.11 Rearview mirrors. An adjustable rearview mirror not less than 90 square inches shall be mounted on each side of the truck near the windshield. The mirrors shall be so positioned that the driver can clearly observe the roadway and vehicles approaching from the rear on either side.

3.10.12 Windshield wipers and washer. Windshield wipers (two or four as required) shall be mounted to facilitate clear vision. The wipers shall be of such capacity and coverage as to readily cleanse the windshield of rain, snow, ice, and foam when used in conjunction with the defrosting system or windshield washer mechanism. The wiper arms and blades shall be of sufficient length to clear at least 75 percent of the windshield area. Wiper assemblies shall include features to permit speed control and return of the wiper blades to a retracted position out of the line of vision when not in use. The windshield cleaning system shall include a device that will spray the path of the wiper blades with a water or water alcohol cleansing solution. The cleansing solution reservoir shall have a minimum capacity of 2 gallons. The system shall be pressurized with air to produce a high volume blast of spray. In addition, a cooling spray from the agent system, controlled by a separate valve, shall be provided to keep the windshield cool when the truck is exposed to radiant heat from fires.

3.10.13 Reflectors. Reflectors conforming to MIL-R-12144 shall be mounted on the truck or body. Four amber reflectors shall be installed at the front, two visible from the front and one visible from each side. Four red reflectors shall be installed at the rear, two visible from the rear and one visible from each side.

3.10.14 Forward view mirrors. A mirror at least 60 square inches shall be provided on each side of the cab at the top of the windshield. The mirrors shall be

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so installed that the driver and equipment operator can see personnel, equipment, or fires directly in front of the vehicle that otherwise would be obscured from sight below the bottom of the windshield as covered in 3.5.4.2 and figure 1.

3.10.15 Clothing storage. Space equivalent to 12 by 18 by 36 inches for the storage of firemen's clothing shall be furnished within the body of the vehicle and accessible to the crew during travel.

3.10.16 Rear platform. Platform space(s) equal to at least 16 by 16 inches for each of at least two crew members shall be provided at the rear of the truck for use as standing room during structural fire fighting operations. The platform space(s) shall have nonslip steel plate tread surface(s) and shall be located not more than 24 inches above ground level. If necessary to meet specified ram clearance requirements, the platform space(s) may be hinged at the forward edge(s) to permit folding upward. If hinged, means shall be provided to selectively hold the platform space(s) in the uppermost and lowered position. Platform space(s) shall have sufficient strength to support personnel weighing 250 pounds each without permanent deformation during off-road service. Handrails fabricated from 1-1/4 inch minimum diameter chromium-plated brass, chromium-plated steel, or polished stainless steel tubing shall be provided for each crew member intended to occupy rear platform space(s) during off-road service. The handrails shall assist personnel stepping onto the platform space(s) and provide the required stability for crew members while occupying the platforms. The handrails shall be readily accessible to personnel with physical dimensions in the range of small to large as defined in SAE J833. The handrails shall be so mounted as not to interfere with operations or removal of hose and equipments.

3.10.17 Hose beds. Hose beds shall be provided with adequate capacity for separate storage of 500 feet of 2-1/2 inch and 300 feet of 1-1/2 inch, double jacketed, cotton, rubber-lined fire hose in horizontal layers. The beds shall be designed and mounted to permit hose loading without restrictions. The mounting shall be such as to permit both normal maintenance accessibility to vehicle components, and withdrawal of the hose to the rear of the truck while making a hose lay. The mounting shall incorporate provisions for leading the hose and preventing damage to truck components during hose withdrawal operations. Leading hose sections shall be accessible to crew personnel while standing on the rear platform space(s) (see 3.10.16). The hose bed interiors and hose leading provisions shall be free of all projections, sharp or rough edges which might chafe or tear fire hose, catch hose couplings, or hinder hose placement or removal. Removable slatted flooring constructed of well-seasoned, select grade oak or other hardwood having similar characteristics shall be provided for the hose beds. The slats shall be not less than 3-1/2 inches wide and 3/4 inch thick carried on transverse, hardwood sleepers. The slats shall run the entire length of the hose beds, shall be spaced between 5/8 and 3/4 inch apart, and shall be restrained to prevent warping. The ends of the slats shall be rounded to prevent damage to the hose.

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3.11 Instruments and controls. Except where otherwise specified herein, all installed equipment shall be controllable from the cab. Instruments and controls shall be located within the limits specified in SAE J898 and grouped according to applicability and relative function. Switches and other controls having similar functions shall have similar actions. Instruments and controls shall be so arranged that those which are most vital to operation will have the most predominant location. They shall be placed in blocks distinctly marked to group those pertinent to the operation of any individual component. For example, all instruments and controls relative to the operation of the engine shall be in one block. Instruments and controls related to the operation of the truck proper shall be readily accessible to the driver. Instruments and controls related to the operation of fire fighting equipment and agents shall be readily accessible to both the equipment operator and the driver from their respective sitting positions. Indicator lights shall be green to indicate proper functioning, and red when warning is intended.

3.11.1 Subpanels. Instruments and controls which are provided for kit operations shall be grouped into subpanels. These subpanels shall be removable and a blank panel provided when the kits are not installed.

3.11.2 Panel instruments. Twenty-four-volt instruments shall be properly installed on the instrument panel unit, readily removable for servicing purposes or rear access provisions incorporated, and located for effective use by the driver and fire fighting equipment operator, respectively. At least the following instruments shall be included on the panel:

a. Speedometer and recording odometer

b. Engine tachometer

c. Ammeter

d. Fuel gage

e. Engine temperature gage

f. Oil pressure gage

g. Engine oil temperature gage

h. Transmission oil temperature gage

i. Air pressure gage

j. Fire pump pressure gage

k. Water tank temperature gage

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l. Booster heater tell-tale light

m. Booster heater cycling light

n. Coolant pump pressure gage.

3.11.3 Panel lighting. The instrument panel shall include adequate nonglare lighting to properly illuminate all instruments, switches, and controls.

3.11.4 Panel controls. At least the following controls shall be included on the instrument panel, except when a more suitable location is approved:

a. Master power switch

b. Engine switch (no key required)

c. Starting switch

d. Hand throttle

e. Light switches

f. Windshield wiper and washer controls

g. Defroster control (if applicable)

h. Heating and ventilating

i. Booster heater switch

Water recirculating pump switch (three-position; off, cycling, run)

3.11.5 Tank level gages. Tank level gages shall be provided for both the water and foam concentrate tanks. The gages shall be mounted inside the cab in a readily visible location to both the driver and equipment operator. The gages shall be of the type that incorporate a manual air pump mounted on the gage and utilize air pressure to determine head in the tank. The gages shall allow a determination of quantity of agent in the tanks with an accuracy of ±5 percent of the individual tank capacities at any level from empty to full.

3.12 Agent system. The agent system shall include the agent tanks, pump inlet piping, water and foam concentrate pumps, pump discharge piping, agent proportioning system, agent discharge turrets and hose reel, control system, drain system, flushing system, and system control panel. The equipment operator's seat shall be immediately to the right of the driver in the cab.

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3.12.1 Agent system valves. Unless otherwise specified, agent system manual valves shall be of the quarter turn full-on to full-off type. All valves shall be mounted to be readily removable without disturbing other piping and components. All valves shall drain free in order to seal and be operable without damage under sub-freezing conditions as specified herein. All wearing parts shall be replaceable or adjustable.

3.12.2 Agent system piping. The agent system piping and all components shall withstand a hydrostatic pressure of 350 psi on the suction side of the pumps and 500 psi on the discharge side of the pumps without leaking. All piping sections between valves and other agent system components shall be readily removable without disturbing the components. Seamless tubing and groove-type couplings shall be used to the maximum practical extent. Large pipe sections and pipe sections which could be subjected to excessive movement during operation shall be firmly mounted or braced. Sharp bands and changes in cross section shall be avoided. The piping shall be designed and mounted to facilitate drainage. Piping wall thickness shall be a minimum of schedule 40 for all materials. All pipe openings from heated compartments to the outside of the truck shall be protected with grommets or other devices to limit air passage around the pipes.

3.12.2.1 Hose construction. Hose conforming to MS28741 may be used in the piping system in lieu of metal pipe and tubing. Hose sections of more than 1-1/8 inch ID shall not be used. Hose sections shall be neatly dressed and fastened to structure and routed so as not to interfere with operation or maintenance of the system.

3.12.3 Agent tanks. The water and foam concentrate tanks may be either separate or integral tanks. The tanks shall be mounted to be removable by no more than disconnecting piping to the tanks, removal of bolt-on outer truck sheet metal, and unbolting the tank mountings. The tanks shall be provided with removable top covers to allow inspection and access for cleaning and repair of all portions of the tanks. Each tank shall incorporate a hinged or easily removable inspection cover at least 15 inches in diameter. The top covers shall be sealed to prevent leakage under any operational condition. Connections to the tanks shall be reinforced and provided with flexible joints or hose to allow for motion of the tank and piping. The tanks shall be provided with longitudinal and transverse baffles to limit rapid shifting of the agent. Both longitudinal and transverse baffles shall be no more than 3 feet on center in the tanks. Baffles or other tank interior structure shall prevent vortices during pumping. The tanks shall be fabricated of 300 series stainless steel. High quality fiberglass reinforced plastic materials may be used in the fabrication of the tanks when approved by the procuring activity. Plastic materials shall conform to L-P-383 and the process specification shall be submitted for approval. The external surfaces of the tanks, if used as external body surfaces, shall be flat and smooth. The water and foam concentrate tanks shall withstand an internal static pressure of 5 psi without leaking.

3.12.3.1 Agent tank vent and overflow systems. The water tank shall incorporate at least a 3-inch diameter pipe overflow system. The foam tank shall incorporate

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at least a 1-inch diameter pipe overflow system. The overflow systems will relieve excess liquid when the tanks are being filled. Each tank shall incorporate a vent system which may be combined within the overflow system. The vent systems shall provide for relieving positive and negative pressure on the tanks during tank fill and agent system discharge. The overflow and vent system shall prevent leakage of water and foam liquid from the tanks when filled to capacity and operating on the maximum side slopes and grades and mobility conditions specified herein and during all agent system pumping operations. The vent and overflow systems shall not be affected by any of the environmental conditions specified herein including freezing conditions. Drainage from the vent and overflow systems shall not flow over body panels or other truck components. Pipe or hose extensions shall be provided so all points of discharge will be below the chassis frame. The foam tank vent shall be directed downward into the filler box to recover any loss during filling due to foaming.

3.12.3.2 Agent tank drains. The agent tanks shall incorporate drain pipes and shutoff valves. The drain valves shall be controlled by an operator at the left (driver's) side of the truck. The points of discharge of all drains shall be below the chassis frame. The foam tank drain shall be located so the contents of tank can be drained into 5 gallon cans. The water tank drain shall be at least 3 inches in diameter. The foam tank drain shall be at least 1 inch in diameter.

3.12.3.3 Water tank. The water tank shall have a capacity of at least 1,500 gallons. The tank shall be designed and mounted so the center of gravity of the tank will be centered on the longitudinal axis of the truck for any load condition on level terrain. The tank shall be mounted on the chassis frame and the center of gravity of the fully loaded tank shall be no more than 24 inches above the top of the frame rail. The tank shall be so designed that at least 90 percent of its contents can be discharged at the maximum rated capacity of the agent system (1,200 gpm) when operating in any direction on a 20-percent slope.

3.12.3.3.1 Water tank filler box. A filler box at least 10 by 10 inches shall be provided in the top of the tank readily accessible to an operator on top of the truck. The filler box shall incorporate a hinged cover and latch. The filler box shall be designed to allow a 2-1/2-inch hose coupling to be placed in the opening and the tank to be filled at any flow rate up to 350 gpm without being held by the operator. The filler box shall incorporate a removable strainer to prevent entry of foreign matter into the tank during all filling operations. The strainer shall be rigidly constructed and at least 10 mesh.

3.12.3.3.2 Water tank loading connection. The tank shall incorporate a 2-1/2 inch male NH hose connection terminating at the rear of the truck to permit filling from a hydrant or other vehicle at flow rates up to 350 gpm. The connection shall be provided with a chained-on rocker lug cap. The fill line shall be mounted to preclude accumulation of water and freezing at low temperatures. The connection shall be no higher than 48 inches from the ground and readily accessible. The fill line shall either fill the tank through the filler box or incorporate a check valve and readily removable, rigidly constructed 10 mesh strainer.

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3.12.3.4 Foam concentrate tank. The foam tank shall have a capacity equal to 12 percent of the water tank capacity to the nearest 10 gallons. The center of gravity of the tank shall be centered on the longitudinal axis of the truck when in the filled condition. The tank shall be so designed that 90 percent of the contents can be discharged at the rated capacity of the agent system (72 gpm) when operating in any direction on a 20-percent slope. The tank shall be mounted away from the truck engine and other heat producing components and protected from radiation in order to minimize heat input. The tank may be adjacent to or integral with the water tank.

3.12.3.4.1 Foam tank filler box. The foam tank shall incorporate a filler box mounted in the top of the tank readily accessible to at least two operators on top of the truck. The filler box shall be at least 6 inches high and incorporate a hinged cover and latch to prevent spillage under any operating condition. The filler box shall be designed to be filled by an exterior transfer hose of 1-1/2-inch diameter at any flow rate up to 75 gpm. The tank shall incorporate a readily visible marker to indicate when it is filled to capacity. The filler box shall be capable of allowing four standard 5-gallon foam liquid containers to be emptied simultaneously. Can puncturing devices set into the filler box shall be provided for the four cans. The puncturing devices shall allow for drainage of each container in 20 seconds. The filler box filler neck shall extend near enough to the bottom of the tank to reduce foaming to a minimum during the filling operation. The filler box shall incorporate a removable strainer to prevent entry of foreign matter into the tank during all filling operations. The strainer shall be rigidly constructed and at least 10 mesh.

3.12.3.4.2 Foam tank loading connection. The tank shall incorporate a 1-1/2-inch NH male hose connection terminating at the rear of the truck to permit filling by an exterior transfer hose at flow rates up to 75 gpm. The connection shall be provided with a chained-on pin or rocker lug cap. The fill line shall be mounted to preclude accumulation of foam and freezing at low temperatures. The connection shall be no higher than 48 inches from the ground and readily accessible. The fill line shall fill the tank through the filler box.

3.12.4 Pump inlet piping. Inlet piping for the water and foam pumps shall provide for continuous flow of liquid to the pumps without cavitation at all flow rates up to rated system capacity. Tank shutoff valves shall be provided in the inlet piping of both the water and foam pumps. The water tank shutoff valve shall be controlled at the left (driver's) side of the truck at or near the control panel. The foam tank shutoff valve shall be controlled from the equipment operator's seat and driver's seat in the cab. The foam tank shutoff valve shall be located to minimize wetted piping. The water pump inlet piping shall incorporate a check valve to prevent backflow into the tank when the tank shutoff valve is open and the truck is connected to a hydrant. The foam tank inlet piping shall include a check valve to prevent backflow under any condition.

3.12.4.1 Hydrant connections. The water pump inlet piping shall include a 5-inch NH suction connection with long handled cap on each side of the truck and a 2-1/2-inch

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NH gated, gasketed, swivel female suction connection with chained-on pin or rocker lug cap on each side of the truck. The connections on the left (driver's) side shall be located on the control panel and the connections on the right side in the same relative position. The connections shall be no more than 4 feet above the ground. All hydrant connections shall incorporate removable strainers in each inlet. Strainers shall be rigidly constructed, precision casting or all welded, with approximately 1/4 inch openings and 3/8 to 1/2 inch deep. All caps shall be brass or chromium-plated brass. The connections shall be used for pumping from draft or from a hydrant. Piping to the connections shall be sized corresponding to the size of the connection without the use of reducers.

3.12.5 Water pump. The unit shall incorporate a single-stage centrifugal pump. The suction inlet of the pump shall be lower than the bottom of the water tank to assure gravity flow to the pump under all conditions specified herein. The pump shall be removable as an assembly without disturbing other components. Mechanical pump seals, if provided, shall be replaceable without removal or disassembly of the pump. The pump shall be so constructed and mounted that water will not enter into any gearbox or transmission in the event of pump seal failure.

3.12.5.1 Pump shaft and impeller. The pump shaft shall be [illegible]ated of stainless steel. The impeller shall be brass or bronze.

3.12.5.2 Pump drive system. The pump shall be driven by the truck engine through a clutch controllable from the cab by the driver and equipment operator in their normal seated positions. The clutch shall provide for smooth engagement and disengagement of the wet pump with the engine operating at any speed up to maximum governed speed and with the truck drive system in any operating condition.

3.12.5.3 Priming pump. The priming pump shall be of the electric motor-driven positive displacement type complete with lubricating oil reservoir. The priming pump shall be controlled by a combination switch and priming valve control mounted on the outer control panel. The priming pump and motor assembly shall be so mounted that it can be easily serviced and removed without disturbing other components. The motor shall be so mounted that it can be removed without removing the priming pump. The lubricating oil reservoir shall be at least 6 quart capacity and allow syphoning of oil to the priming pump when the priming pump is being operated. The tank shall incorporate a dipstick and drain. The tank shall be mounted in a side compartment where it can be easily filled and the oil level checked by an operator on the ground at the side of the truck. Access shall be provided for draining the tank.

3.12.6 Foam concentrate pump. The pump shall be either a centrifugal or positive displacement type driven by the truck engine through a clutch controllable from the cab by the driver and the equipment operator in their normal seated positions. The clutch shall provide for smooth engagement and disengagement of the pump with the engine operating at any speed up to maximum governed speed and with the truck

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drive system in any operating condition. The suction inlet of the pump shall be lower than the bottom of the foam tank to assure gravity flow to the pump under all conditions specified herein. The foam concentrate pump drive shall be so arranged that the water pump must be operating any time the foam concentrate pump is operating.

3.12.7 Agent discharge piping. The discharge piping shall incorporate the water pump pressure relief bypass system, the pumper discharge connections, and the foam concentrate pump bypass system. Both the water and foam discharge piping shall prevent entrance of water into the foam concentrate piping and components, except during flushing (see 3.12.13) and foam into the water piping and components upstream of the proportioning system under all conditions specified herein.

3.12.7.1 Water pump pressure relief bypass. The pressure relief bypass shall incorporate a pump to water tank return line, pump bypass, and control valves. The pressure relief bypass shall insure that proper system pressure is maintained for operation of the turret and handline nozzles at rated capacity in any on-off combination within 3 seconds after any nozzle is turned on or off with any engine throttle setting above that required for maximum agent system performance. The pressure relief bypass control shall automatically engage from the equipment operator's position for operation of the turrets and handline overriding the outer control panel for both water and foam discharge. The pressure relief bypass system shall allow for bypass of water to the tank during crash fire fighting operation, but not during pumper operation. The pressure relief bypass system shall be controllable for pumper operation from the outer control panel with a hand wheel to allow any desired pressure setting from 100 to 300 psi. The pump bypass shall limit pump discharge pressure to no more than 500 psi at rated pump speed against a totally closed discharge when the hand wheel is adjusted for maximum pressure.

3.12.7.2 Pumper discharge connections. Two 2-1/2-inch NH male discharge connections shall be located on the left (driver's) side of the truck and two on the right side in the same relative position. The discharge connections shall be mounted at a 45° downward angle. Each connection shall incorporate a valve of the quarter turn quick closing ball type with the twist to lock in any position. The valves on the right shall be operable from either the right or left side of the truck. The connections shall be no more than 48 inches above the ground. Each connection shall be provided with a chained-on pin or rocker lug cap. All caps shall be brass or chromium plated brass. Piping to the connections shall be sized corresponding to the size of the connections without use of reducers.

3.12.7.3 Foam concentrate pump pressure relief bypass. The pressure relief bypass shall incorporate a pump to foam concentrate return line and control valve(s). The return line shall pass through the water tank to allow for cooling if the foam pump is operating against a closed discharge. The pressure relief bypass shall insure that proper system pressure is maintained for operation of the turret and handline nozzles at rated capacity in any on-off combination within 3 seconds after any nozzle is turned on or off with any engine throttle setting above that required for maximum agent system performance.

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3.12.8 Proportioning system. The proportioning system shall automatically and uniformly proportion water and foam concentrate at a ratio of 6 parts of foam concentrate to 94 parts water by volume with a tolerance of 0 to ±10 percent. The proportioning system shall be of the pressure balancing type and automatically provide the foam as specified above to the roof turret at both the 400 and 800 gpm flow rates, the bumper turret at rated capacity, and the handline at rated capacity either individually or simultaneously in any combination.

3.12.9 Agent discharge turrets. Roof and bumper turrets shall be installed at the front of the cab. The installation shall provide for frontal approach to the fire, but shall permit agent application to either side of the truck. All nozzles and piping thereto, from the valves out, shall be vented and automatically drain when the valves controlling discharge to the nozzles are in the off position to provide fast draining to prevent freezing during low temperature operation. Means shall be provided to prevent the turrets from inadvertently leaving their storage positions due to truck motion yet not restrict instantaneous employment when desired. Each turret shall be controllable by the equipment operator and the driver in the normal sitting positions. All outside control cables shall be protected from the weather and shall be self-lubricating. Nozzle barrels may be fabricated of aluminum.

3.12.9.1 Roof turret. The roof turret shall be of the double-barrel, single nozzle type (two foam and two water) mounted on the cab roof. The nozzle shall be capable of discharging either foam or water selectively from one or both barrels. Each barrel shall have a capability of discharging at least 400 gpm (water rate) to produce a total single nozzle discharge capacity of at least 800 gpm (water rate) when both barrels are operated at the same time.

3.12.9.1.1 Roof turret directional control. The roof turret shall be controlled by a single control lever to provide movement of the nozzle in the horizontal and vertical planes through similar displacements of the control lever which corresponds in azimuth and elevation with the position of the turret. The control system shall incorporate hydraulic power assist. The hydraulic power assist system shall be independent from any other hydraulic system installed on the truck. The maximum force required to move the control lever shall not exceed the limits specified in MIL-STD-1472 for one hand lever operation. The horizontal arc of straight stream agent coverage shall be at least 210° symmetrical about the centerline of the truck. The vertical arc of straight stream agent coverage shall extend from the position where the nozzle just clears the truck to an elevation of at least 60° above the horizontal. The turret shall depress to a position to be capable of providing a foam coverage exceeding 0.20 gpm per square foot on an arc no more than 50 feet from the truck throughout the full horizontal movement of the nozzle.

3.12.9.2 Bumper turret. The bumper turret shall be of the single barrel, single nozzle (one water and one foam) type mounted in front of the cab as low as possible but out of the angle of approach in the stowed position. The turret shall be capable of discharging either foam or water selectively at a rate of at least 300 gpm (water rate).

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The turret shall provide ground sweep protection by initial positioning of the turret. The ground sweep position of the turret shall provide a foam agent coverage with low velocity ground impingement exceeding 0.20 gpm per square foot for the maximum dispersion pattern of the nozzle at a distance of no more than 20 feet from the front of the truck. The ground sweep position and maximum dispersion pattern of the nozzle shall be maintained while allowing operator on-off selection of the turret discharge control.

3.12.9.2.1 Bumper turret directional control. The bumper turret shall be controlled by a single control lever to provide movement of the nozzle in the horizontal and vertical planes through similar displacements of the control lever which corresponds in azimuth and elevation with the position of the turret. The turret shall be manually controlled. The maximum force required to move the control lever shall not exceed the limits specified in MIL-STD-1472 for one hand lever operation. The horizontal arc of straight stream agent coverage shall be at least 170° symmetrical about the centerline of the truck. The vertical arc of straight stream agent coverage shall include an arc from at least 45° above the horizontal to a depressed position capable of providing a foam agent coverage exceeding 0.20 gpm per square foot on an arc no more than 20 feet from the truck throughout the full horizontal movement of the nozzle.

3.12.9.3 Turret discharge controls. Each turret shall incorporate an on-off discharge control capable of being operated by the equipment operator in the normal sitting position and easily accessible to the driver. The roof turret discharge valve shall be power actuated with a manual override to allow operation in the event of power failure. The roof turret directional control lever shall incorporate a trigger or pushbutton to interrupt the turret discharge and resume discharge when the trigger or pushbutton is released. In the event of failure, the valve shall remain in the open position.

3.12.9.3.1 Nozzle and agent selection control. Each turret shall have separate nozzle dispersal pattern controls, water-foam selection controls and the roof turret shall have a 400/800 gpm flow control. The controls shall be operable by the equipment operator in the normal sitting position and easily accessible to the driver. The nozzle dispersal pattern controls shall provide for any position from straight stream to the widest dispersal pattern position for foam discharge. The nozzle dispersal pattern controls shall provide at least straight stream and dispersed (water fog) pattern for water. Once the control positions are selected, they shall remain positioned and not be affected by operation of any other controls. The water-foam selection controls shall incorporate lock-in features to preclude accidental change of selection. The nozzle dispersal pattern controls shall be capable of operation during agent discharge.

3.12.9.4 Turret nozzle performance. The turret nozzles shall selectively discharge water and foam in a straight stream and dispersed patterns. All

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discharge patterns shall be homogeneous, symmetrical, and contiguous. The nozzles shall produce a water application rate of at least 0.20 gpm per square foot over the pattern area at the maximum pattern dispersal when the turret is elevated for maximum range.

3.12.9.4.1 Water discharge. The turret nozzles shall discharge water in a dispersed pattern of conically shaped flow with very minutely sized droplets. The included angle of the cone shall be at least 45°. The distance from the tip of the nozzle to the maximum diameter of the cone projection on the ground shall be at least 50 feet. When discharging in straight stream, the distance from the tip of the nozzle to the far side of the discharge pattern where the application rate is at least 0.20 gpm per square foot shall be at least 150 feet.

3.12.9.4.2 Foam discharge. The turret nozzles shall discharge foam with at least eight expansion and having a 25-percent drainage time above 4-1/2 minutes when measured as specified in NFPA Standard 412 using foam concentrate in accordance with O-F-555. The turret nozzles shall discharge foam with at least five expansion and having a 25-percent drainage time above 4-1/2 minutes when measured as specified in NFPA Standard 412 using foam concentrate in accordance with MIL-F-23905. The turret nozzles shall be capable of this performance at any pattern dispersal when the turret is elevated for maximum range. The spread of the dispersed patterns of the turret nozzles and the minimum distance from the tip of the nozzles to the far side of the patterns when the turrets are elevated for maximum range and where the application rate is at least 0.20 gpm per square foot shall be as specified below:

Turret	Straight Stream Distance (Feet)	Dispersed Pattern Distance (Feet)	Dispersed Pattern Width (Feet)
Bumper	125	60	25
Roof (both rates)	160	80	30.

3.12.10 Foam handline. A combination water foam agent handline shall be provided. The handline shall be capable of selectively discharging either foam at a rate of at least 100 gpm (water rate) or water at a rate of 50 gpm. The handline shall incorporate a type II nozzle conforming to MIL-N-26206 and 150 feet of 1-1/4 inch ID hose. The hose conforming to MIL-H-4497 shall be in 50-foot sections joined by reusable couplings conforming to MIL-C-7580. The nozzle shall be capable of producing foam of at least eight expansion and having a 25-percent drainage time above 4-1/2 minutes when measured as specified in NFPA Standard 412 using foam concentrate in accordance with O-F-555. The nozzle shall be capable of producing foam of at least five expansion and having a 25-percent drainage time above 4-1/2 minutes when measured as specified in NFPA Standard 412 using foam concentrate in accordance with MIL-F-23905. The nozzle attached to the handline shall be stored in a compartment at the right front corner of the truck and shall be capable of being withdrawn

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by an operator standing at the side of the truck. Also included shall be a bayonet nozzle so fabricated that it can be used in conjunction with the assembly of the handline nozzle. The bayonet nozzle shall be mounted in a quick-release bracket either in the handline nozzle compartment or in a separate protected location adjacent to the handline nozzle compartment. The bayonet nozzle shall connect into the bayonet joint of the handline nozzle assembly. Quick attach-detach provisions shall be incorporated for installing the bayonet nozzle.

3.12.10.1 Hose reel. The handline hose shall be mounted on a 150-foot capacity reel which shall be located in a compartment which provides for handline hose withdrawal to the front of the truck. Horizontal and vertical rollers shall be provided to reduce chafing and friction of the hose or catching of the couplings when the hose is being withdrawn to the front or either side. Provisions shall be incorporated to permit flushing of the handline hose and reel compartment. The design of the hose reel assembly shall be such as to prevent dragging, snarling, and overrunning of the reel during unreeling of the hose and agent discharge operations. Manual means for rewinding the hose which will include features to eliminate rewind motor drag shall be provided. Electrical means for rewinding the hose at an average rate of 85 fpm shall also be provided. The rewind control switch shall be positioned to allow one person to actuate the switch and feed the hose on the reel. The switch shall be protected from hose interference. Controls for operation of the handline valve shall be placed in or adjacent to the handline nozzle compartment. It shall not be necessary to clean the hose or nozzle after use at the scene of an emergency in order to accomplish rewind operations and storage. An air purge system, with check valves to prevent water from entering the truck air system, shall be provided to purge the hose to prevent freezing under low temperature conditions. The air purge system control shall be located in the handline nozzle compartment.

3.12.11 Control system. The control system shall include all manual and automatic controls required for operation of the agent system as specified herein. Automatic controls shall be pneumatic, hydraulic, or electrically operated. The controls shall be arranged to allow full operation of the agent system for crash fire fighting by the equipment operator in a normal seated position. Driver access to controls shall be as specified herein. Operation of the truck as a structural fire pumper shall be controlled by an operator standing on the ground at the control panel on the left (driver's) side of the truck. The agent system shall exhibit no instability in pressure or flow rate under any operating condition.

3.12.11.1 Pressure surge. The control system shall incorporate surge arresting devices to prevent pressure surges under any condition which could cause damage to the truck after numerous occurrences. The control system shall be capable of allowing simultaneous shutoff of the turret nozzles and the handline without exceeding the normal governed speed of the engine.

3.12.11.2 Control system maintenance adjustment. The control system shall be so designed and installed that all control adjustments can be accomplished in a specified

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sequence, and all adjustments are independent. The control system shall be so designed that all adjustment points are accessible without the removal of any parts or components.

3.12.12 Drain system. In addition to the agent tank drains specified herein, the agent system including all components shall incorporate drains to provide for rapid and complete draining of the system. Once drained, the truck shall withstand temperatures down to -40°F without causing any damage to the agent system or any of its parts and components. Careful attention shall be given to draining gage and control line tubing to prevent freeze damage. Drainage shall not be allowed to flow over or onto other truck components. Rubber hose at least the size of the drain valve shall be provided to conduct the flow to the outside of the truck enclosure. Drain valves which are not readily accessible from beside the truck shall be provided with remote controls which will permit draining without crawling under the truck. Drain valves shall be grouped to reduce the number of locations. Drain lines shall be grouped to reduce the number of valves. In addition to the agent tank drains, no more than six drain control locations shall be required to completely drain the system. The drains shall be sized so the entire agent, other than the agent tanks, can be drained in no more than 4 minutes.

3.12.13 Flushing system. The agent system shall incorporate a flushing system to rapidly flush all foam residue out of the agent piping from the foam tank shutoff valve to the agent discharge turrets and handline. The flushing system shall be controlled by the equipment operator in the normal sitting position.

3.12.14 Agent system control panel. An agent system control panel for operation of the truck as a structural fire pumper shall be provided on the left (driver's) side of the truck. All controls and instruments shall be arranged in patterns and grouped depending on function for maximum convenience of the operator. Switches and other controls having similar functions will have similar and corresponding actions. All controls shall be operable by personnel wearing heavy gloves. All controls shall be spaced to prevent interference and be firmly supported. The control panel shall be located on the outside of the truck within easy reach and clearly visible to an operator standing on the ground. The control panel shall be easily accessible from the rear for maintenance. All instruments and controls shall be illuminated with nonglare lighting to be readily visible for night operation. The control panel shall include at least the following instruments and controls:

a. Pump pressure gage (30 inches Hg vacuum to 600 psi)

b. Pump suction gage (30 inches Hg vacuum to 400 psi)

c. Water pump relief valve control

d. Priming pump control

e. Engine tachometer (at least 4 inches in diameter)

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f. Engine throttle - Pull type with pushbutton latch and turn for microadjustment

g. Engine oil pressure

h. Engine coolant temperature

i. Panel light controls

j. Hydrant connections

k. Discharge connections

l. Discharge valve controls.

3.12.14.1 Pressure gages. Pressure gages shall be at least 4-1/2 inches in diameter and flush mounted type. Each gage shall be provided with a throttle valve and control located directly beneath the gage to allow elimination of pressure pulsations. The gages shall have face-mounted adjustment and removable front glass. The gages shall have no less than 20 psi graduations. The gages shall have a 2-1/2 percent accuracy of the full scale range over the full scale range with the vacuum and pressure ranges considered individually. Each gage line shall incorporate a 1/4-inch NPT plugged test connection on the control panel.

3.12.14.2 Engine throttle interlock. An interlock shall be provided to prevent operation of the agent system control panel engine throttle except when the truck transmission (drive line) is in neutral, and to prevent shifting the transmission into gear when the control panel engine throttle is advanced beyond the engine idle position.

3.13 Auxiliary fire fighting equipment. Necessary compartments, holders, toolboxes, brackets, and mounting pads shall be provided on all trucks to accommodate the auxiliary equipment specified below. Such brackets, holders, etc, shall be so designed and installed that the equipment will be held firmly in place under all operating conditions and readily removable for use. The following items shall be provided on the truck:

a. One axe conforming to GGG-A-926, type II, mounted in a storage compartment

b. Two 2-1/2-inch NH straight double female hose connectors conforming to MIL-C-52404, class A, type XVIII, stored in a storage compartment

c. One coupling, gated 2-1/2-inch NH internal by 2-1/2-inch NH external, conforming to MIL-C-52404, type XVII, class C

d. Two extinguishers conforming to MIL-E-7589; one mounted on each side of the truck

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e. One ladder conforming to MIL-L-12197, class 3, style B stored in a suitable compartment

f. One crowbar conforming to GGG-B-101, type II, class 1, size 4, mounted in a storage compartment

g. One claw-type door opener conforming to GGG-B-101, type V, class 1, style B, mounted in a storage compartment. Lightweight metals may be used to reduce the weight of the tool

h. Two adjustable hydrant and spanner wrenches conforming to GGG-W-665, type IV, stored in a storage compartment

i. One MXU-346/E circular blade metal cutting saw conforming to MIL-S-38265 stored in a storage compartment

j. An electric lantern conforming to MIL-L-26598, class W, mounted in the cab. The lantern shall have an adjustable shoulder strap to permit a crew member to carry it and have both hands free

k. One pike pole conforming to MIL-P-43116, type III, style A, size 1, except it shall be storable in the ladder compartment. The pike pole handle shall be at least 7 feet long

l. One bolt cutter conforming to GGG-C-740, type II, class E, 5/8-inch rod stored in a storage compartment.

3.14 Fire fighting kit, structural KMK-391/S32P-4. Necessary compartments shall be provided on all trucks to accommodate the components indicated for compartment storage. Mounting pads or bases for these items shall be drilled and tapped or otherwise prepared for attachment-detachment of the necessary holders or brackets.

3.14.1 Kit components. The following components of the KMK-391/S32P-4 kit with necessary holders, brackets, and attaching devices shall be furnished either installed on the truck or as a separate package as specified (see 6.2).

a. Five hundred feet of 2-1/2-inch fire hose (50-foot lengths) conforming to ZZ-H-451, type III (hose bed storage)

b. Three hundred feet of 1-1/2-inch fire hose (50-foot lengths) conforming to ZZ-H-451, type III (hose bed storage)

c. Three 7-foot sections of 5-inch woven wire filler, or braided wire filler, helix construction hard suction base with 5-inch NH long handle couplers on the female ends and 5-inch NH rocker lug couplers on the male ends (compartment storage)

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d. Two 2-1/2-inch NH straight double male hose connectors conforming to MIL-C-52404, type XIX (compartment storage)

e. One siamese connection, 2-1/2-inch NH female swivel inlet, two-gated 1-1/2-inch NH male outlets conforming to MIL-C-52404, type XIV, class B (compartment storage)

f. Two nozzles, 2-1/2-inch NH adjustable fog to straight stream, conforming to MIL-N-12314, type II, class A, with provisions for selecting the stream before opening the valve (compartment storage)

g. Two nozzles, 1-1/2-inch NH adjustable fog to straight stream, conforming to MIL-N-12314, type II, class A, with provisions for selecting the stream before opening the valve (compartment storage)

h. Two reducers, 2-1/2-inch NH female gasketed rocker lug coupling to 1-1/2-inch NH male end conforming to MIL-C-52404, type XV, class C (compartment storage)

i. One strainer conforming to MIL-S-12165, type I, with 5-inch NH female couplings (compartment storage)

j. One double internal hose reducer, 5-inch NH by 4-1/2-inch NH, conforming to MIL-C-52404, type XV, class A (compartment storage)

k. Two nipples, 1-1/2-inch NH external conforming to MIL-C-52404, type XIX (compartment storage)

l. Two 1-1/2-inch NH straight double female hose connectors conforming to MIL-C-52404, type XVII, class A (compartment storage)

m. Two universal hose coupling wrenches conforming to GGG-W-665, type V (compartment storage)

n. Six hose and ladder straps conforming to MIL-S-11086 (compartment storage)

o. One lever-operated hose clamp for 2-1/2-inch single jacketed hose (compartment storage)

p. One hose jacket conforming to MIL-J-449 (compartment storage)

3.15 Winterization. The winterization system specified herein shall permit satisfactory operation of the truck and fire fighting systems and provide required heating in ambient temperatures as low as -40°F. The winterization system shall be powered by the truck electrical system and shall be designed to provide the required performance during all phases of truck and fire fighting operations regardless of truck

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engine speed. The system shall also provide the required performance when the truck engine is shut off and the vehicle battery charge is being maintained from an external 110V ac power source. The system design and installation shall include the necessary coolant flow shut-off features to permit removal and reinstallation of the major components of the system without requiring the draining of coolant from the system and loss of coolant in excess of the capacity of the component being removed. The winterization system shall be so installed and the vehicle so insulated that, after stabilization of operating temperatures, the system when operating in temperatures as low as -40°F, can be shutdown for a period of at least 2 hours without requiring the draining of the agent system and without freeze damage to the truck and its components. Incorporation of the winterization features shall not detract from the performance of the truck or fire fighting systems in normal or high ambient temperatures up to 125°F. The winterization system shall include but not be limited to the following features:

a. Coolant circulating pump

b. Booster heater

c. Cab and compartment heating and ventilating system

d. Agent system heat exchangers

e. Water recirculating pump

f. Engine heating system

g. Control system

h. Auxiliary power provisions.

3.15.1 Heat circulation and distribution. The winterization system shall provide for forced flow of coolant solution starting at an electrically driven coolant circulating pump and continuing through a closed circuit which includes at least the following: the booster heater, the defroster and cab heater cores, finned tubing for compartment heating, heating pads (if required) for the agent system pumps and foam metering valves, the agent system heat exchangers, and the oil pan and cooling system of the engine. The coolant distribution system shall be provided with an expansion tank, filler opening, and drains so located as to be readily accessible for inspection and servicing. The capacity of the expansion tank shall be adequate to assure that all coolant lines will remain full if the coolant system is allowed to cool to a temperature of -40°F.

3.15.2 Electrically driven coolant circulating pumps. The electrically driven coolant pump shall be of the centrifugal type designed for 24V dc potential. The

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rated capacity of the pump shall be the flow through the booster heater that is required to maintain a rise across the heater of at least 30°F when using winter coolants. The pump shall be equipped with roller or ball type bearings. The pump shall be capable of satisfactory operation at -40°F. A relay shall be provided which will blow the vehicle horn in event of coolant circulating system failure.

3.15.3 Booster heater. A booster heater of at least 90,000 BTU per hour output shall be provided in the heating system. It shall operate on diesel fuel conforming to VV-F-800 and 24V dc electrical potential. With the exception of the use of diesel fuel, the heater shall conform to the requirements of MIL-H-4697.

3.15.3.1 Automatic shutoff. Automatic switches shall be provided to shutoff the heater in the event of ignition and overheating failures. A relay shall be provided which will blow the vehicle horn in the event of heater failure.

3.15.3.2 Booster heater installation. The booster heater shall be installed in a compartment in an accessible location which will permit servicing and removal of the units. The supplier shall provide all necessary controls not incorporated in the heater, and the signaling features specified in 3.15.3.1. The booster heater shall be mounted upon a sliding carriage which will permit easy individual removal of the unit from the compartment.

3.15.3.3 Heater exhaust. The booster heater exhaust shall be so located and installed as to prevent damage to any equipment due to radiation or conduction of heat, entrance of exhaust fumes into the cab, and the entrance of reverse-drafts into the exhaust pipe which would cause malfunctioning of the heater.

3.15.3.4 Booster heater drains. Drain lines shall be provided to drain the heater water jackets to the outside of the vehicle.

3.15.4 Cab and compartment heating and ventilating system. The design and installation of the cab heating system shall incorporate at least air heater cores, electrically driven air blowers, dampers to accomplish the required mixing or excluding of inside and outside air inducted into the blower, ducting to direct a flow of heated air over transparent areas, and variable speed controls.

3.15.4.1 Cab heating and ventilating system. The cab heating and ventilating system shall induct at least 15 cfm of fresh air into the crew space per crew member and shall maintain a temperature of at least 40°F at the floor of the cab when the outside temperature is -40°F. With the cab being occupied by four members, the defrosting and anti-icing provisions shall be capable of keeping at least 75 percent of the windshield and each of the other transparent areas free of fog on the interior surfaces and ice on the exterior surfaces of the windshield when the outside temperature is -40°F. Ducts shall be provided with control dampers to facilitate balancing of the flow in the lines and to permit adjustment during operations according to requirements.

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3.15.4.2 Compartment heaters. Compartments shall be heated by finned tubing. The heat output of these heaters shall be adequate to maintain the temperature at the floor of the compartments at 40°F when the outside temperature is -40°F.

3.15.5 Agent system heat exchangers. The agent system heat exchangers shall provide adequate heated coolant flow to maintain required tank temperatures in ambient temperatures as low as -40°F.

3.15.5.1 Water system heat exchanger. The water system heat exchanger installation shall include all valving and connections necessary to induce flow of water from the bottom of the water tank and return through all wetted piping to the water tank. When the main valve is open, the water flow shall be through the suction piping, water recirculating pump, heat exchanger, agent water pump, discharge piping, and return to the water tank. When the main valve is closed, the water flow shall be from the suction pipe (near the main valve), through the water recirculating pump and heat exchanger to the water tank. Valves for the required flow shall be mechanically interconnected so as to reduce the number of controls to the minimum. This system shall be capable of heating the water in the tank regardless of the water level. The return line from the discharge lines to the water tank shall be provided with check valves to prevent discharge to the tank from these lines when the fire pump is in operation.

3.15.5.2 Water recirculating pump. An electrically driven water pump designed for 24V dc potential shall be installed to recirculate water through all wetted piping.

3.15.6 Agent pump heaters. If required, removable and installable heater pads may be provided which will attach to the agent pump impeller cases through which coolant shall flow for heating of the pumps.

3.15.7 Foam system heaters. If required, removable and installable heater pads or jackets may be provided which will attach to the foam system components for heating with coolant flow from the heating system.

3.15.8 Engine heating system. The engine heating system shall provide a flow of heated coolant through the oil pan and cooling system of the vehicle engine. The design shall be such as to assure the standby and starting performance specified without affecting the engine's normal cooling and lubricating systems. The heating system shall be adequate to maintain required engine temperatures to assure engine starting within 15 seconds in temperatures as low as -40°F.

3.15.8.1 Shutoff valve. A manual shutoff valve shall be provided which will isolate the truck engine coolant system from the coolant flow in the winterization circulating system.

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3.15.9 Control system

3.15.9.1 Cab heating and ventilating system controls. Heating and ventilating system controls shall be so installed on the cab control panel that they are readily accessible to the driver and equipment operator.

3.15.9.1.1 Defroster control. Rheostats shall be furnished to control the speed of blowers for defrosting, anti-icing, and heating and ventilating. The necessary air mixing valve controls may be located apart from the control panel, but shall be accessible either to the driver or the equipment operator.

3.15.9.2 Water recirculating pump switch. A 3-position control switch shall be provided for the water recirculating pump. The center position shall be the "off" position; the upper position shall be the "on-cycling" position which will actuate the pump when the temperature in the lines drop below 45°F and stop the pump when the line temperature approaches 70°F; the lower position of the switch shall be the "on-run" position and shall run the pump regardless of water temperature. Thermal control switches shall be placed in the normally coldest lines or locations. The required number of thermal switches necessary for satisfactory sampling of temperatures shall be incorporated. A relay shall be provided which will blow the vehicle horn in event of water recirculating system failure.

3.15.9.3 Booster heater tell-tale-light. A red bead light shall be installed on the control panel and connected to the booster heater automatic shutoff switch specified in 3.15.3.1 to indicate when the booster heater has failed.

3.15.10 Auxiliary power provisions. The necessary connectors, transformers, rectifiers, overload circuit breakers, etc, shall be properly installed in the truck and connected in parallel with the voltage regulator and reverse current relay of the truck electrical system to maintain the truck battery charge from an external 110V, 20 amp, 60 Hz ac power source while the winterization system is operating and the truck engine is not running.

3.15.10.1 Electrical cable. A 50-foot electrical cable with connectors conforming to style X of W-C-596 shall be provided for connecting the truck to the external power source. The cable shall be equipped with a male connector at one end and a female connector at the other end. The cable shall be two No. 10 conductor plus ground and shall conform to either MIL-C-3432, class HO, flexibility F, or to type SO commercial cable. The cable shall be stored in a compartment on the truck when not in use.

3.15.10.2 Truck receptacle. A recessed male receptacle conforming to style X of W-C-596 shall be so mounted at the rear of the truck that the cable will pull loose without damage to the truck or the cable if the truck is driven away while the cable is connected. The power terminals of the receptacle shall be connected to the battery charging system and the grounding terminal shall be connected to the chassis. The receptacle shall have a spring-loaded, weatherproof cap included as part of its housing.

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3.16 Air conditioning kit. When specified (see 6.2), the truck shall be provided with an air conditioning kit. The kit shall consist of the cab air conditioning system and the required ducting and mounting devices for installing the kit. The kit installation shall provide accessibility for routine servicing and preventative maintenance without removal from the truck.

3.16.1 Cab air conditioning system. The truck cab shall be cooled by a standard automotive-type air conditioning system. The air conditioner shall be designed for use with dichlorodifluoromethane (refrigerant 12). The air conditioner shall be sized to reduce the cab temperature to 85°F within 30 minutes and maintain the temperature while the truck is in a ready standby condition with engine at idle speed and exposed to 115°F ambient temperature. Flexible transitions shall be provided between the air conditioner evaporator supply and return air openings and the truck cab. The air conditioner shall be capable of operating in the normal cooling mode (not on bypass) at all engine speeds within the range of ambient conditions described on a psychrometric chart by the following limits:

a. The 40°F dry bulb line to intersect with its line of saturation

b. The saturation line from 40° to 84°F

c. A line from 84°F saturation point to 97°F dry bulb and 87°F wet bulb

d. A line from 97°F dry bulb and 87°F wet bulb to 125°F dry bulb and 75°F wet bulb

e. The 125°F dry bulb line to 125°F dry bulb and 75°F wet bulb.

3.16.1.1 Air conditioning compressor. An automotive type compressor powered by the truck engine shall be provided. The compressor shall be of sufficient capacity to insure compliance with the cooling requirements specified herein.

3.16.1.2 Air conditioning condenser. The condenser shall be an aircooled type with tube and fin construction. The condenser shall withstand 1-1/2 times the saturated refrigerant pressure at 160°F without rupture or distortion.

3.16.1.3 Air conditioning evaporator. The evaporator shall be of tube and fin construction and shall withstand an internal pressure of 1-1/2 times the saturated refrigerant pressure at 160°F without rupture or distortion. The evaporator shall have adequate surface and tube distribution to insure the cab temperature requirements specified in 3.16.1.

3.16.1.3.1 Evaporator fan. The evaporator fan shall be sized and selected to insure compliance with the cooling and noise requirements specified herein. The fan shall be driven by a multispeed motor that obtains power from the vehicle electrical system.

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3.16.1.3.2 Evaporator drain. The evaporator shall be equipped with a drip pan and drain for disposal of condensate water overboard.

3.16.1.4 Controls. The air conditioning system shall incorporate safety controls for protection of the system in accordance with SAE J639, and shall have adequate controls for starting the air conditioner and controlling air temperature from within the truck cab.

3.16.1.5 Piping. Refrigerant piping shall be sized to provide minimum pressure drops throughout the system. Maximum practical use shall be made of flexible refrigerant lines and quick-disconnect couplings between major components.

3.16.1.6 Noise level. When measured as specified in 4.6.1.20.4.2, the noise level generated by the air conditioner kit shall not exceed the following values:

Octave Band Center Frequencies (Hz)	Octave Band Pressure Level (In Decibles Ref. 0.0002 Microbar)
63	85
125	80
250	76
500	73
1,000	71
2,000	70
4,000	69
8,000	68.

3.17 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.18 Electromagnetic interference. The truck shall comply with the requirements of MIL-STD-826, method 7001. This requirement shall apply to the truck, including any and all equipment used in or on the truck, and during operation under any conditions specified herein.

3.19 Dimensions. The truck shall conform to the following dimensions:

a. Width - 96 inches maximum

b. Height - The overall height of the empty truck in a fully operational configuration shall not exceed 132 inches. The overall height of the empty truck shall be reducible as required for air transport (see 3.5.7)

c. Length - Minimum practicable.

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3.20 Curb weight. The curb weight of the truck shall not exceed 28,200 pounds. The curb weight includes the truck with all attachments, accessories, kits, and equipment with full complement of lubricants and coolants and 15 gallons of fuel. Curb weight does not include crew and fire fighting agents. This shall be the truck weight in the air transport configuration (see 3.5.7).

3.21 Finishes and protective coatings. The truck shall be treated in accordance with MIL-STD-808 and painted as follows: Exterior surfaces shall be finished in full gloss red, color 11136, conforming to Standard 595. The interior surfaces of the cab and compartments shall be green, film designation CS, in accordance with MIL-STD-808.

3.22 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following special characteristics shall be added:

Capacity:

Water: _____ (Gallons)

Foam Concentrate: _____ Gallons)

Make and Model _____

Vehicle Weight, Unloaded _____

Gross Weight, Maximum _____

Date of Delivery _____

Fuel Grade _____

Oil, Engine, Above 32°F, SAE Grade _____

Oil, Engine, Below 32°F, SAE Grade _____

3.22.1 Mounting of nameplate. The nameplate containing the information specified in 3.22 shall be mounted in the driver's compartment and shall be the sole source of information on the vehicle as to its make and type. Except for the serial number and date of delivery, which may be stamped, all data shall be inscribed.

3.23 Markings

3.23.1 Instruction and data plates. Transportation data plates sized and marked as required in MIL-P-514 shall be provided. All other data plates, placards, charts, instruction plates, etc, shall conform to MIL-P-514 of a size and shape consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12. Symbols shall be in accordance with MIL-STD-17.

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3.23.2 Operational markings. The operational markings and stenciling shall be applied in accordance with the marking requirements of MIL-STD-808.

3.23.2.1 Tire pressure. The tire inflation pressure shall be stenciled on the truck as near each wheel as practicable.

3.23.3 Identification of controls. All valves, gages, controls, and indicators, other than standard vehicular instrumentation, used in the operation of the truck shall be identified by securely attached nameplates of such composition that exposure to oil, dirt, light, et cetera, will not cause them to fade or become eradicated. Tags or decals shall not be used.

3.23.4 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed on or near the controls and switches. The instructions shall be clear, concise, and adequate to enable operation without damage to the equipment or injury to personnel.

3.23.4.1 Caution plates. The following caution plates shall be installed in the cab on the instrument panel. The caution plate pertaining to engine operation shall be located adjacent to the tachometer.

IF THE ENGINE IS OPERATED FOR ANY LENGTH OF TIME AT A SPEED GREATER THAN _____ RPM AS SHOWN BY THE ENGINE TACHOMETER ON THE INSTRUMENT PANEL, THE ENGINE WILL BE SERIOUSLY DAMAGED.

WARNING - BUZZER OPERATION IS AN INDICATION OF LOSS OF BRAKE AIR PRESSURE.

CAUTION

THIS VEHICLE IS EQUIPPED WITH RADIO INTERFERENCE SUPPRESSION DEVICES. CARE SHOULD BE EXERCISED DURING VEHICLE MAINTENANCE TO RETAIN AND REINSTALL THESE DEVICES PRIOR TO REUSE OF VEHICLE.

3.23.4.2 Draining instruction plate. An instruction plate shall be installed in the cab containing specific data concerning draining of the cooling system, agent tanks, pump, and lines.

3.23.4.3 Flushing instruction plate. An instruction plate shall be installed in the cab containing specific data concerning flushing of the foam/water system piping.

3.23.4.4 Lubrication plate. A lubrication plate shall be provided directing attention to all lubrication fittings and shall specify the range and grade of lubricant required for operational temperatures. The plate shall be permanently attached to the trucks in an accessible and convenient location.

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3.23.4.5 Pump delivery instruction plate. An instruction plate shall be installed at the pump control compartment indicating the water pump delivery rate in gallons per minute at 100, 150, 200, 250, and 300 psi pressures with the truck manufacturer's specified engine speeds for these delivery rates.

3.24 Workmanship. The truck, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, marking of parts and assemblies, thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws, bolts, etc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the truck shall be classified as follows (see 6.2):

4.2.1 Preproduction testing	See 4.4
a. General verification and performance testing	See 4.6.1
b. Truck mobility and performance testing	See 4.6.2
c. Operational testing	See 4.6.3.
4.2.2 Acceptance tests	See 4.5
a. Individual tests	See 4.5.1
b. Sampling plan and tests	See 4.5.2.

4.3 Test conditions

4.3.1 Preparation for test. All trucks submitted for test shall contain all specified components and shall be completely assembled and properly prepared for operation. The trucks, including all components, shall be correctly adjusted, lubricated, and serviced; and all equipment which has been ordered shall be mounted for immediate operation. Fuel and agent tanks shall be full unless otherwise specified for a particular test.

4.3.2 Test fluids. Unless otherwise specified, the test fluids shall be tap water and foam concentrate in accordance with O-F-555. The temperature of the fluids shall not differ more than 5°F and shall be within the temperature range of 45° to 70°F.

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4.3.3 Data. During the tests specified herein, at least the following data, unless not applicable, shall be recorded at intervals not greater than 15 minutes or as otherwise indicated. Additional data may be required and recording intervals decreased for test of short duration or as otherwise dictated by test objectives.

a. Data

b. Time started

c. Time finished

d. Ambient temperature

e. Atmospheric pressure

f. Agent level in the water tank

g. Water tank temperature

h. Agent level in foam concentrate tank

i. Foam concentrate tank temperature

j. Water pump inlet pressure (in Hg)

k. Water pump discharge pressure

l. Foam concentrate pump discharge pressure

m. Agent flow rate through each nozzle

n. Engine speed (rpm)

o. Engine oil temperature

p. Engine water temperature

q. Torque converter oil temperature.

4.3.4 Test observations. Throughout all tests specified herein, the truck shall be closely observed for the following conditions which shall be considered cause for rejection:

a. Failure to conform to performance requirements specified herein

b. Spillage or leakage of any liquids including fire fighting agents, coolants, and lubricants under any operating condition

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c. Structural failure of any part including permanent distortion, or evidence of impending failure

d. Evidence of excessive wear

e. Interference between truck components or between the truck and the ground and all required obstacles with the exception of normal contact by the tires

f. Misalignment of parts and components

g. Instability in agent system pressure or flow

h. Evidence of undesirable roadability features including instability in handling during turning, braking, and when traversing rough terrain

i. Conditions which present a safety hazard to personnel during operation, service, or maintenance

j. Overheating of the truck engine or any components of the truck.

4.4 Preproduction testing

4.4.1 Test samples. Unless otherwise specified in the contract (see 6.2), one fully equipped truck including all kits and attachments shall be subjected to all tests specified under 4.6.

4.4.2 Test report. Upon completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831.

4.4.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions at the time of the events identified shall be recorded.

b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure or irregular functioning.

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made or proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such corrections.

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d. Clock time and manhours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions.

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item.

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.4.2.2 Cleaning and finishing procedures. A description of all cleaning and finishing procedures in accordance with MIL-STD-808 to be used on the production trucks and all components shall be included as an appendix or attachment to the test report.

4.4.2.3 Component test reports. Test reports on all components covered by detail specifications requiring tests under the applicable specification shall be included as an appendix or attachment to the test report.

4.5 Acceptance tests

4.5.1 Individual tests. Each truck shall be subjected to the following tests as described under 4.6:

a. Examination of product 4.6.1.1

b. Features and components 4.6.1.2

c. Agent tanks, static pressure test 4.6.1.3

d. Agent system, hydrostatic pressure test . . . . . 4.6.1.4.

4.5.2 Sampling plan and tests. One truck shall be selected at random from each 50 or fraction thereof produced and subjected to the following tests as described under 4.6:

a. One-hundred-mile road test 4.6.1.5

b. Cross-country 4.6.1.6

c. Agent system foam discharge . . . . . 4.6.1.14.

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4.5.2.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all tests influenced by the failed components shall be repeated.

4.5.2.2 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a test failure. Final acceptance of the items on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 General verification and performance testing

4.6.1.1 Examination of product. The truck shall be inspected to determine compliance with requirements of this specification with respect to materials, dimensions, workmanship, and markings.

4.6.1.2 Features and components. All features and components shall be inspected for proper installation, adjustment, and serviceability. The truck shall be driven for a distance of at least 5 miles over paved highway and an average speed of 40 mph and the agent system, including all nozzles individually and in all combinations, shall be operated while the truck is in motion. All components shall be moved through their maximum normal displacement during applicable operations of driving and agent system discharge, and shall exhibit no malfunctioning, leaks, objectionable rough operation, or irregular movement.

4.6.1.3 Agent tanks, static pressure test. The water and foam concentrate tanks shall be subjected to an internal static air pressure of 5 psig for not less than 30 minutes. A soap solution shall be sprayed on the exterior of the tanks to facilitate detection of leaks that may exist. Evidence of air leakage shall be cause for rejection.

4.6.1.4 Agent system hydrostatic pressure test. The agent system piping and all components on the suction side of the pumps from the tank shutoff valves shall be subjected to a hydrostatic pressure of 350 psi for 15 minutes. The agent system piping and all components on the discharge side of the pumps shall be subjected to a hydrostatic pressure of 500 psi for 15 minutes.

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4.6.1.5 One-hundred-mile road test. The truck, with agent tanks 75 percent full, shall be operated over hard surface highway for a distance of 5 miles at 60 mph and for a distance of 95 miles at an average speed of 45 mph. During the test, the following functions shall be performed:

a. Normal or average stops, 10 times

b. Quick stops from a speed of at least 20 mph by instantaneous application of full service braking power, five times

c. Sharp right turns and sharp left turns, five times each

d. Operation through all forward and reverse gears including the transfer case, if applicable.

4.6.1.6 Cross-country. The truck, with agent tanks 75 percent full, shall be operated over cross-country terrain (see 6.3.1) for a distance of at least 30 miles. Differential locking devices shall be engaged for at least 1 mile of operation.

4.6.1.7 Rotating beacon. The rotating beacon shall be placed in a properly instrumented chamber and operated to demonstrate compliance with the characteristics specified in 3.9.3.

4.6.1.8 Ramp test. The truck with gross load shall be driven forward from a solid, flat, horizontal surface, and ascend a solid ramp equal in length to 1.2 times the wheelbase of the truck and making an angle of 30° with the horizontal. It shall proceed across the ramp crest onto an upper connecting horizontal surface until all wheels are resting on the upper surface. The truck shall then be backed from the upper horizontal surface, down the ramp and onto the lower horizontal surface. The truck shall be turned around and backed up the ramp onto the upper horizontal surface and then driven forward down the ramp and onto the lower horizontal surface.

4.6.1.9 Water tank top fill. The empty water tank shall be filled with water from a 2-1/2-inch hose placed in the filler box at an average flow rate of at least 350 gpm. The quantity of water required to fill the tank shall be measured.

4.6.1.10 Water tank rear loading. The empty water tank shall be filled with water from a 2-1/2-inch hose using the loading connection at the back of the truck at an average flow rate of at least 350 gpm.

4.6.1.11 Foam concentrate tank top fill. The empty foam concentrate tank shall be filled with foam concentrate from a 1-1/2-inch hose placed in the filler box at an average flow rate of at least 75 gpm. The quantity of foam concentrate required to fill the tank shall be measured.

4.6.1.12 Foam concentrate tank rear loading. The empty foam concentrate tank shall be filled with foam concentrate from 1-1/2-inch hose using the loading connection at the back of the truck at an average flow rate of at least 75 gpm.

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4.6.1.13 Foam concentrate tank fill (5 gallon cans). It shall be demonstrated that four 5-gallon cans of foam concentrate can be emptied into the tank by two men in no more than 20 seconds using can puncturing devices installed in the filler box.

4.6.1.14 Agent system foam discharge (O-F-555 foam concentrate). The truck shall be operated to demonstrate that the agent system will produce foam from the roof turret at a water flow rate of at least 400 gpm (single barrel) and 800 gpm (double barrel), the bumper turret at a water flow rate of at least 300 gpm, and the handline at a water flow rate of at least 100 gpm, individually and in all combinations, using foam concentrate in accordance with O-F-555. The test shall be conducted for each nozzle individually and in all combinations for all nozzles in straight stream pattern, then repeated for all nozzles in the widest dispersed pattern. Maximum pattern dimensions and ranges, based upon a minimum of 0.20 gpm per square foot, shall be determined. The foam discharge from each nozzle individually and each nozzle in combination shall be analyzed to verify that the foam concentrate is between 6 and 6.6 percent of the total volume of agent. The foam discharge from each nozzle individually and each nozzle in combination shall be analyzed to determine foam quality; at least 8 expansion and having a 25-percent drainage time above 4-1/2 minutes when measured as specified in NFPA Standard 412. The agent system shall be observed to determine that there is no instability in pressure or flow rate under any operating condition and that each initial flow condition and all new nozzle flow combinations are achieved within 3 seconds. The same truck engine throttle setting shall be maintained for all portions of this test. Each flow condition shall be maintained for a minimum of 30 seconds. The roof turret flow interrupt control shall be operated at least three times to demonstrate satisfactory operation. The roof turret shall be operated to determine that it can discharge agent exceeding 0.20 gpm per square foot on an arc no more than 50 feet from the truck throughout the full horizontal movement of the nozzle. The test shall be repeated for the bumper turret, except the distance shall be no more than 20 feet. Foam agent coverage will provide low velocity ground impingement.

4.6.1.15 Agent system foam discharge (MIL-F-23905 foam concentrate). The truck shall be operated to demonstrate that the agent system will produce foam from the roof turret at a water flow rate of at least 400 gpm (single barrel) and 800 gpm (double barrel), the bumper turret at a water flow rate of at least 300 gpm and the handline at a water flow rate of at least 100 gpm using foam concentrate in accordance with MIL-F-23905. The tests shall be conducted for each nozzle individually in straight stream pattern, then repeated for each nozzle individually in the widest dispersed pattern. Maximum pattern dimensions and ranges, based upon a minimum of 0.20 gpm per square foot, shall be determined. The foam discharge from each nozzle shall be analyzed to verify that the foam concentrate is between 6 and 6.6 percent of the total volume of agent. The foam discharge from each nozzle shall be analyzed to determine foam quality; at least 5 expansion and having a 25-percent drainage time above 4-1/2 minutes when measured as specified in NFPA Standard 412. The agent system shall be observed to determine that there is no instability in pressure or flow rate under any operating condition. The same truck engine throttle setting shall be maintained for all portions of this test. Each flow condition shall be maintained for a minimum of 30 seconds.

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4.6.1.16 Agent system water discharge. The truck shall be operated to demonstrate that the agent system will discharge water from the roof turret at a flow rate of at least 400 gpm (single barrel) and 800 gpm (double barrel), the bumper turret at a flow rate of at least 300 gpm, and the handline at a flow rate of at least 50 gpm, individually and in any combination. The tests shall be conducted for each nozzle individually and in all combinations for all nozzles in straight stream pattern, then repeated for all nozzles in the widest dispersed pattern. Maximum pattern dimensions and ranges, based upon a minimum of 0.20 gpm per square foot, shall be determined. Included cone angles of at least 45° shall be measured for turrets in the maximum dispersed pattern position. The agent system shall be observed to determine that there is no instability in pressure or flow rate under any operation condition. The same truck engine throttle setting shall be maintained for all portions of this test. Each flow condition shall be maintained for a minimum of 30 seconds.

4.6.1.17 Stationary pumping test. The truck shall be operated as a structural fire pumper. The truck shall discharge water at a rate of at least 1,000 gpm at a pump discharge pressure of 150 psig while drafting at a 9- to 10-foot lift through 20 feet of 5-inch hard suction hose for a period of at least 2 continuous hours. The truck shall be completely drained at the beginning of this test and the priming pump used to initiate flow. The maximum vacuum during priming shall be recorded. The truck shall then be adjusted to pump at approximately 100, 200, 250, and 300 psi. Each pressure-flow condition shall be maintained for at least 15 minutes. The level of water in the reservoir shall be maintained at the 9- to 10-foot lift level throughout the entire test.

4.6.1.18 Dry pump operation. The agent system shall be completely drained. The water and foam pumps shall be operated at rated pump speed for 10 continuous minutes. The agent tanks shall then be filled and the system operated and checked for leaks. If there is evidence of damage, such as abnormal noise, pumps shall be disassembled and inspected.

4.6.1.19 Agent tank exhaustion. The capability to discharge 97 percent of the water tank capacity by normal means when the truck is on level ground and is headed up, down, right, and left on a 20-percent slope shall be demonstrated.

4.6.1.20 Environmental tests. During the environmental operational tests, temperatures at least at the following points shall be recorded at intervals not greater than 15 minutes:

a. Ambient - four dispersed places

b. Coolant inlet and outlet at booster heater

c. Coolant inlet for all major components and compartments being heated

d. Air at center and bottom of all compartments being heated

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e. Air in cab at air inlet, foot levels, center, and ceiling

f. Windshield at center and all four corners

g. Agent in center of agent tanks

h. Water at recirculating pump, critical piping locations, and all nozzle shutoff valves

i. Foam concentrate at pump outlet, critical piping locations, and all inductors.

4.6.1.20.1 Low temperature storage. An empty truck, processed for storage, shall be placed in a cold chamber and subjected to a soak period of at least 72 hours after temperature stabilization within the chamber at -80° ±2°F has been obtained. After the soak period, the chamber shall be warmed to existing ambient temperature and the vehicle thoroughly inspected for damage. If there is no apparent damage, the vehicle shall be serviced, the agent tanks filled to 75 percent full, and the truck driven for at least 25 miles on paved highway. The speed shall be varied from 10 mph to maximum with the average being 30 mph. During this road test, the following operations shall be performed at least five times each: panic stops, jack-rabbit starts, and right- and left-hand turns at a speed of 20 mph. After the paved highway operation, agent shall be discharged simultaneously at rated capacity through all the various nozzles. At least one complete filling of the agent tanks shall be discharged.

4.6.1.20.2 Low temperature operation. The truck shall be serviced with the proper low temperature greases, oils, and winter coolants; filled with fuel and agents. The agent system shall be drained and the drain time recorded. The truck shall be instrumented in accordance with 4.6.1.20 and the system inspected for undrained pockets of agent. The truck shall be placed in a cold chamber using refrigerated air and the temperature within the chamber lowered and maintained at 0° ±2°F for a period of at least 24 hours or until temperature stabilization is reached. The capability to start the engine at 0°F without use of the winterization system shall be demonstrated. The agent tanks shall be filled; the truck connected to draw power from an external 110V, 20 amp, ac power source; the winterization system turned on and set for automatic operation; and the air temperature within the chamber lowered and maintained at -40° ±2°F for a period of at least 24 hours or until temperature stabilization is reached. During the period that the winterization system is being operated from an external power source, critical temperatures (see 4.6.1.20) shall be recorded. Water shall be sprayed on the transparent areas of the truck at the close of the period to demonstrate the capability of the anti-icing system. After the period of operation from an external power source and with the chamber temperature maintained at -40° ±2°F, the capability of the truck engine to start within 15 seconds at -40°F shall be demonstrated. After the truck engine has been started, the external power shall be disconnected, and the winterization system operated from vehicle power for at least 8 hours with the vehicle engine at idle speed to simulate vehicle standby operation. Critical temperatures shall be recorded during the period. At the

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start of the period, all agent dispensing nozzles shall be brought into operation momentarily and the handline purged using the installed air purge system. The momentary dispensing of agent from all nozzles and air purge of the handline shall be repeated after 2 and 6 hours of the simulated standby operation. Evidence of freezing shall be noted. A crew of four persons shall occupy the cab for a period of at least 4 hours during the simulated standby period of operation. Cab heating, defrosting, and anti-icing features shall be tested. A small amount of agent shall be drained from each drain in the agent system, once during the first hour and again during the last hour of simulated standby operation. Evidence of freezing shall be recorded. After the simulated standby operation and with the chamber temperature maintained at -40° ±2°F, the vehicle engine and the winterization system including water circulating pump, shall be shutoff for a period of 2 hours. Critical temperatures shall be recorded and evidence of freezing noted.

4.6.1.20.3 High temperature storage. An empty truck, properly processed for storage shall be placed in a hot chamber and subjected to a soak period of at least 72 hours after temperature stabilization within the chamber at 160° ±2°F has been obtained. After the soak period, the chamber shall be cooled to existing ambient temperature and the vehicle thoroughly inspected for damage. If there is no apparent damage, the vehicle shall be serviced, the agent tanks filled to 75 percent full, and the truck driven for at least 25 miles on paved highway. The speed shall be varied from 10 mph to maximum with the average being 30 mph. During this road test, the following operations shall be performed at least five times each; panic stops, jack-rabbit starts, and right- and left-hand turns at a speed of 20 mph. After the paved highway operation, agent shall be discharged simultaneously at rated capacity through all the various nozzles. At least one complete filling of the agent tanks shall be discharged.

4.6.1.20.4 High temperature operation. The truck shall be filled with fuel, instrumented in accordance with 4.6.1.20 (except for winterization coolant system temperatures), placed in a hot chamber, and subjected to an ambient temperature of 125° ±2°F for a period of at least 24 hours, or until temperature stabilization is reached. The truck engine shall be started, agent tanks filled, and all component equipment, including the air conditioning kit, checked to determine that they will operate satisfactorily at the 125°F temperature. Agent dispensing equipment shall be tested by discharging the agent simultaneously at rated capacity through the various nozzles. At least one complete filling of the agent tank shall be discharged. With the ambient temperature maintained at 125° ±2°F, the fire pump shall be operated to discharge 1,000 gpm of water at 150 psi continuously for a 5-hour period to assure proper truck engine cooling capacity when the truck is used as a pumper.

4.6.1.20.4.1 Air conditioner test. After high temperature operation at 125°F, the temperature of the chamber shall be lowered to 115° ±2°F and maintained for a period of at least 24 hours, or until temperature stabilization is reached. The air conditioner shall be started. During this test, a crew of four shall occupy the truck

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or their metabolic heat load simulated. The cab temperature shall be lowered to 85°F and the time to reach this temperature recorded. The 85°F cab temperature shall be maintained for 2 hours. At the end of this test, all safety controls shall be checked for effectiveness. The air conditioner shall exhibit no malfunctioning, leaks, rough operation, vibration, or other irregularities as a result of this test.

4.6.1.20.4.2 Noise level test. The criteria specified in 3.16.1.6 are levels for the operator and crew positions. The levels are measured at head level in the cab by slowly moving the measuring microphone in a circle 1 foot in diameter at these positions. The average level for each octave band shall be determined by arithmetic average of the minimum and maximum noise levels (in db) found. Measurements shall be made with the evaporator fan operating at the maximum speed. All acoustic measurement equipment shall meet the requirements specified in USA Standard S1.4 and Z24.10, except the octave band analyzer shall employ the preferred frequencies specified in Standard S1.6.

4.6.1.20.5 Other environmental tests. The components and materials used in the construction of the truck shall either be subjected to or certified as having passed or be capable of passing the following environmental tests as described in MIL-STD-810:

a. Rain - Method 506, procedure I

b. Humidity - Method 507, procedure I

c. Fungus - Method 508, procedure I

d. Salt fog - Method 509, procedure I

e. Dust - Method 510, procedure I.

4.6.1.20.5.1 Other environmental test evaluation. A critical evaluation of the fully assembled truck shall be conducted to demonstrate that the completed truck is capable of required performance when subjected to the environmental conditions specified in 4.6.1.20.5. The following data shall be approved by the procuring activity prior to evaluation (see 6.2):

a. Name and qualification of person(s) making evaluation

b. Time and place of evaluation

c. Criteria to be considered during evaluation.

4.6.1.21 Air transportability tests

4.6.1.21.1 Configuration testing. The reduced height for air transportability shall be measured after removal of demountable units. If a removable cab roof

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is provided, it shall be tested for compliance with the sealing requirements after 25 complete assembly and disassembly operations using stored kit equipment. Data shall be submitted in accordance with the Alternate load tests specified in MIL-A-8421.

4.6.1.21.2 Loading demonstration. The truck shall be disassembled to the extent necessary for air transport. The removed portions shall be stored in or on the truck. The truck shall be loaded aboard a C-130 aircraft equipped with an A/A32H-4 dual rail cargo handling system in accordance with the Loading demonstration test specified in MIL-A-8421.

4.6.1.22 Electromagnetic interference. The truck shall be subjected to and shall satisfactorily complete the radio interference suppression tests in accordance with MIL-STD-826, method 7001 when any or all of the equipment used in or on the truck is operating.

4.6.1.23 Reliability demonstration. Compliance with the quantitative reliability requirements shall be considered demonstrated when both the following apply:

a. A reliability prediction, using data, estimates, and procedures acceptable to the procuring activity, will yield a MTBF prediction not less than the specified requirements.

b. All tests specified herein shall be completed with no failure or malfunction that could prevent satisfactory or safe operation in equipment primary functions. Apparent failures that are caused by some unintended factor external to the equipment (eg, test equipment failure) need not be counted as reliability failures if the cause can be substantiated to the satisfaction of the procuring activity. Any test resulting in a failure shall be documented. Servicing, other preventive maintenance, or adjustments authorized by other provisions in this specification need not be counted as failures.

4.6.1.24 Maintainability demonstration. Compliance with the demonstration provisions of the following subparagraphs shall be considered demonstration of compliance with the maintainability and maintenance downtime requirements. Only active maintenance time need be counted. Administrative and supply time need not be counted toward quantitative demonstrations. The capability to perform servicing and maintenance tasks by personnel wearing heavy gloves or mittens shall be demonstrated.

4.6.1.24.1 Preventive maintenance downtime. All preventive and scheduled maintenance tasks shall be done. Tasks that may be termed service shall also be done. The time and the number of men required shall be recorded. As part of this demonstration, the recommended frequencies of the various tasks shall be used with the related recorded times to develop an expected value of preventive maintenance downtime per some measure of use - pumping hours, pumping cycles, or other appropriate operating measure.

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4.6.1.24.2 Corrective maintenance. In demonstrating compliance with the corrective maintenance requirements, relative frequency of corrective tasks may be verified by engineering analysis. This analysis shall be related to the updated failure rates of the reliability prediction, and to the failure mode and effect analysis.

4.6.1.24.3 Corrective maintenance downtime. Compliance with the corrective maintenance downtime requirements shall be demonstrated by a maintenance demonstration procedure, maintenance task selection, and maintenance task performances in accordance with MIL-STD-471, appendices A and B, method 1, test plan pair A1 plus B1. The following additions shall apply to appendix A, columns A and E respectively:

a. Both the estimated task time and estimated task frequency for each item in column A must be identical, otherwise separate entries must be made

b. The percent contribution shall be determined for each item having identical task times and task frequencies by the method outlined.

4.6.1.25 Mechanical checks. A critical inspection shall be made of all components to determine their operability and any damage or undue wear incurred at the end of the preproduction testing program. Teardown and parts measurement shall be made only in those cases where service life is in question.

4.6.2 Truck mobility and performance testing. The following tests shall be conducted at Aberdeen Proving Ground, Maryland:

4.6.2.1 Mechanical inspection. A critical mechanical inspection shall be made of all truck and fire fighting equipment components, and data recorded as to condition; defects of manufacture, damage in transit, and damage through use prior to test.

4.6.2.2 Preliminary operation. If the truck is shipped by rail to Aberdeen Proving Ground, the following operations shall be conducted over level hard-surfaced roads without payload to provide components with necessary operation, under light loads, to seat mating parts and to give operating personnel an opportunity to become familiar with the truck:

a. 50 miles at 25 to 30 mph

b. 100 miles at 35 to 45 mph

c. 150 miles at 5 mph less than maximum speed

If conducted, this operation shall be considered part of the road and cross-country tests.

4.6.2.3 Dimension determination. Measurements for overall length, width, height, wheelbase, and ground clearance shall be made.

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4.6.2.4 Weight determination. The gross and curb weights of the truck shall be determined. Allowance for driver and crew shall be 175 pounds per man.

4.6.2.5 Determination of load distribution. The loads on each wheel and each axle shall be determined at curb and gross weight.

4.6.2.6 Determination of center-of-gravity. The location of the center-of-gravity of the truck at curb and gross weight shall be determined.

4.6.2.7 Suspension flexibility. With the truck at gross load and with the front wheels turned at various angles from full right to full left, opposite wheels, in various combinations, shall be jacked up to allow placement of blocks, equal in height to the cross-section diameter of the tires, under the wheels. Clearances shall be measured and distortion of a permanent nature shall be noted.

4.6.2.8 Determination of turning clearance and turning characteristics. A marker shall be so placed that it will describe the largest circle in a turn at maximum cramping angle. The truck shall then be moved with steering cramped at maximum angle to the left (marker located on the right side) until a circle has been inscribed on the ground. The diameter of this inscribed circle represents the turning clearance to the left. This procedure shall be repeated with the marker on the left side of the truck to obtain the turning clearance to the right. The truck with the agent tanks 75 percent full shall then be subjected to sharp turns at various speeds in both directions and the reaction of the truck observed.

4.6.2.9 Determination of limits of vision. The limits of vision of the driver and equipment operator shall be determined by direct vision of persons, complying with the extremes covered by 3.5.4.2, seated in the truck in their positions and using a large concrete pad marked with concentric circle to indicate distance to right or left of normal. Blind spots within these fields of vision shall be noted.

4.6.2.10 Braking ability. The truck with gross load shall be brought to a stop with the service brakes when headed both up and down a dry, hard surfaced, 60-percent slope. After the truck has been stopped when headed both up and down on a 40-percent slope, the handbrake shall be applied and the service brakes released. The truck shall remain stationary. The truck shall be driven at a speed of 20 mph on level concrete, and the minimum distance required to bring the truck to a stop shall be measured. This shall be accomplished by appropriate methods which will eliminate the reaction time of the driver. The capability of the service brakes to fully recover within 3 minutes at the end of a run down a 3-mile (average 10 percent) grade, with transmission in high gear, and road speed controlled, by repeated application of brakes, to not exceed 30 mph shall be demonstrated.

4.6.2.11 Determination of water-splash characteristics. The distance from the ground to vital parts of the truck which would be affected by short-time (15 seconds)

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exposure to immersion shall be determined by inspection. The ability of the truck to operate at 40 mph in water to a depth of 2 inches shall be determined. After water operation, the time required for recovery of normal braking ability shall be determined.

4.6.2.12 Maximum speed. The truck with gross load shall be operated on dry, level, concrete roadway to determine maximum speed. Not less than 10 test runs in opposite directions shall be conducted. Maximum speed will be the average value obtained.

4.6.2.13 Acceleration. The time-speed relation of the truck with gross load shall be determined while accelerating from a standstill to its maximum speed on level concrete roadway. Not less than 10 test runs in opposite directions shall be conducted. Acceleration will be the average value obtained from each run.

4.6.2.14 Grade speeds. The truck with gross load shall be driven up 5, 10, 15, 20, 30, 40, 50, and 60-percent smooth, dry, hard-surfaced grades and the maximum speed which can be maintained on each shall be determined. Engine oil and fuel pressure shall be observed and engine starting ability shall be determined. The truck shall be driven up and down and stopped when headed in both directions on a 40-percent grade with bumper and roof turrets operated at full discharge to determine if the truck is capable of negotiating the grades without reduction of pump operating pressures.

4.6.2.15 Side slope operation. The 20-percent side slope operational characteristics (both right and left) of the truck with gross load shall be determined upon courses prepared for testing 20-percent side slope operation. Fire fighting equipment engagement and agent discharge characteristics shall be determined during these tests.

4.6.2.16 Fuel consumption. Road-load fuel consumption rates during operation of the truck with gross load shall be recorded. Fuel consumption rates for road and cross-country tests shall be recorded.

4.6.2.17 Cooling. In an ambient air temperature not lower than 60°F, the truck with gross load shall be operated under road-load conditions until normal operating temperatures of the power train components have been reached. Utilizing a mobile dynamometer, the truck shall then be operated for 1 hour in low gear at 70 percent full-load or until one of the component temperatures exceeds these critical values:

a. Engine coolant - 95°F above ambient extrapolated to 125°F

b. Engine oil - 150°F above ambient extrapolated to 125°F

c. Gear box oil - 175°F above ambient extrapolated to 125°F

Temperatures obtained shall be extrapolated on a one-for-one basis for 125°F.

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4.6.2.18 Test courses. The truck shall be tested over prepared and selected test courses to determine the vehicular performance of the test item.

4.6.2.18.1 Vehicle cone index. During 10 separate passes in a different track, the vehicle with gross load shall traverse terrain consisting of unflooded fine grained dry to moist soils having cone index (CI) rating in the critical layer of 50 to 55 (see 6.3.2). During four traversals, the bumper and roof turret shall be operated at full discharge.

4.6.2.18.2 Obstacle crossing ability. The truck with gross load shall be tested over wave course (frame twister) and vertical wall obstacle courses.

4.6.2.18.3 Road and cross-country. The truck shall be subjected to two cycles of operation with the agent tanks full, one cycle of operation with the agent tanks 75 percent full and one cycle of operation with the agent tanks empty over the following roads and terrain at the speeds and distances specified:

Phase	Average Speed (mph)	Maximum Speed (mph)	Distance (miles)
Hard surface highways	40	60	600
Gravel roads	25	35	100
Cross-country level	15	20	100
Cross-country hilly	15	20	100
Belgian block course	15	20	100

During each phase of the above test cycles, the fire fighting equipment shall be engaged and agent discharged from each turret individually and in all combinations. The distance traveled in driving to Aberdeen Proving Ground may be used to satisfy part of the hard-surface highway phases, as applicable, providing complete records are kept during the trip.

4.6.3 Operational tests. Tests shall be conducted to determine the capability of the truck to meet requirements covered in section 3 of this specification under operational conditions (see 6.2).

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with MIL-STD-281, level A, B, or C as specified (see 6.2). Unless otherwise specified, the truck shall be shipped as a mobile unit.

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5.2 Marking. In addition to any special marking required, marking shall be in accordance with MIL-STD-129. The nomenclature shall be as follows:

Truck, Fire Fighting A/S32P-4; Air Transportable, Multipurpose.

6. NOTES

6.1 Intended use. The A/S32P-4 truck is intended to be used as an air transportable, multipurpose fire fighting vehicle for airfield crash-rescue service. When equipped with the KMK-391/S32P-4 structural fire fighting kit it will also provide structural fire service. The truck is intended for use on fixed or forward airbases and may be used either singularly or in conjunction with other fire fighting equipment.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. If a spare wheel and tire assemblies are to be provided (see 3.7.11.4)

c. When the structural fire fighting kit is to be provided and whether it is to be installed on the truck or provided as a separate package (see 3.14.1)

d. When an air conditioning kit is to be provided (see 3.16)

e. Location and conditions for preproduction testing (see 4.2 and 4.6.3) including:

(1) Tests to be conducted by the contractor (see 4.2.1)

(2) Tests to be conducted by or under the direction of the procuring activity (see 4.2.1)

(3) Extent of contractor support and participation during preproduction testing by, or under the direction of the procuring activity (4.2.1)

(4) Number of test samples required (see 4.4.1)

(5) Data to be submitted for approval of environmental test evaluation (see 4.6.1.20.5.1)

f. Level of preservation, packaging, and packing (see 5.1).

6.3 Definitions

6.3.1 Cross-country terrain. Cross-country terrain consists of open fields, broken ground, loose sand or gravel road, and uneven terrain in general.

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6.3.2 Vehicle cone index. Personnel from the Waterways Experiment Station, U.S. Army LCorpos of Engineers, Vicksburg, Mississippi, have empirically developed a relationship between soil types and vehicle mobility with the aid of the cone penetrometer. The procedure for cone penetrometer measurements is reflected in Department of the Army Technical Bulletin TB ENG 37, dated 10 July 1959 and a means of predetermining vehicles mobility characteristics, by computation, is contained in U.S. Army Material Command Technical Memorandum No. 3-240. (18th supplement) dated March 1968. In addition, vehicle cone index requirements have been developed for a large number of representative U.S. Army vehicles. The following definitions pertain to this system:

a. Cone penetrometer - A device used in evaluating trafficability of soils, It consists of a 36-inch long metal shaft with a 30° right circular cone of 0.5 square inch base area mounted on one end of the shaft.

b. Cone index (CI) - A dimensionless value read from the dial gage of the cone penetrometer. The CI is considered an index of the shear resistance of the soil.

c. Vehicle cone index (VCI) - The minimum CI of a soil in the critical layer which will permit a specific vehicle to make 50 passes without immobilization. It can be determined by first computing a mobility index (MI) from an emperical formula given in appendix A of Technical Memorandum 3-240 and referring to a conversion table given in the same appendix.

d. Sheargraph - An instrument that measures surface traction properties of soils at normal pressures equal to the average tire contact pressures on a firm surface.

6.3.3 MTBF. MTBF for reliability purposes is defined as the average (arithmetic mean) operating time between failures that could prevent satisfactory performance of primary functions. Note that accumulated test time without failure equal to the specified MTBF does not demonstrate that specified MTBF at the stated confidence. The full test time required in section 4 is needed.

6.4 Additional aircraft information. Additional information on the Type C-130 aircraft and the A/A32H-4 dual rail cargo handling system is contained in the following USAF cargo aircraft technical orders:

a. T.O. 1C-130A-9 Cargo Loading
C-130A, B, D, E, RC-130A, HC-130B

b. T.O. 1C-130A-9D Supplement to the above

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c. T.O. 1C-130(H) H-9 Loading, Delivery, and Recovery Instructions HC-130H

(Copies of these USAF technical orders should be requested from the contracting officer.)

Custodian:
Air Force - 11

Reviewer:
Air Force - 84

Preparing activity:
Air Force - 11

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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